

Grey-Box Modeling and Application for Building Energy Simulations - A Critical Review

Yanfei Li^{1*}, Zheng O'Neill², Liang Zhang³, Jianli Chen⁴, Piljae Im¹, Jason DeGraw⁵

¹Electrification and Energy Infrastructures Division, Oak Ridge National Laboratory, 1 Bethel Valley Rd, Oak Ridge, TN 37830

²Department of Mechanical engineering, Texas A&M University, 400 Bizzell St, College Station, TX 77843

³Commercial Buildings Research Group, National Renewable Energy Laboratory, 15013 Denver West Pkwy, Golden, CO 80401

⁴Department of Civil & Environmental Engineering, University of Utah, 201 Presidents' Cir, Salt Lake City, UT 84112

⁵Buildings and Transportation Science Division, Oak Ridge National Laboratory, 1 Bethel Valley Rd, Oak Ridge, TN 37830

*Corresponding author: Yanfei Li, liy1@ornl.gov, or Yanfei.tide@gmail.com

This manuscript has been authored by UT-Battelle, LLC, under contract DE-AC05-00OR22725 with the US Department of Energy (DOE). The US government retains and the publisher, by accepting the article for publication, acknowledges that the US government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this manuscript, or allow others to do so, for US government purposes. DOE will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan (<http://energy.gov/downloads/doe-public-access-plan>).

Highlights:

- Grey-box modeling has many advantages within the building energy domain compared with white-box and black-box modeling
- There is no clear naming convention for grey-box modeling structure, regarding the envelope, zone, or whole building
- There are two approaches to create a grey-box model: forward, inverse (data-driven).
- There is a growing interest in grey-box modeling for building applications, including heat dynamic analysis, thermal load estimation, control and optimization, district/urban scale energy modeling, and building-grid integration
- Grey-box modeling has no unified software solution

Abstract

Grey-box modeling, as one of the three fundamental modeling techniques for building energy models, has many advantages compared with black-box modeling and white-box modeling. It has been widely applied to solve problems of building technologies, such as building load estimation, control and optimization, and building-grid integration. However, a thorough review of grey-box modeling is not available. This review study systematically investigated various aspects of grey-box modeling for buildings. First, the fundamental aspects of grey-box modeling are presented, including the theoretical background, modeling of building elements, modeling order, modeling diagram, and order reduction. Second, the detailed modeling approaches are discussed. Third, multiple applications of grey-box modeling are investigated for building energy domain, which are categorized into the following groups: heat dynamics analysis, thermal load estimation, building control and optimization, district/urban scale energy modeling, and building-grid integration. Finally, the available software packages for grey-box modeling are compared. Overall, the challenges of using grey-box modeling can be summarized as follows: (1) the theoretical limitations and assumptions of grey-box modeling are unclear; (2) grey-box model naming convention and structure are confusing; (3) grey-box model creation is vague; (4) suitable applications of grey-box models are unknown; and (5) grey-box models lack unified software solutions for wider adoption.

Keywords: grey-box modeling; resistance-capacitance model; RC model; thermal-network model; simplified model; reduced-order model; hybrid-model; building energy model; heat dynamic analysis; thermal load estimation; building control and optimization; district/urban energy modeling; building-grid integration

Word Count: 8374 (excluding title, author names and affiliations, keywords, abbreviations, table/figure captions, acknowledgments and references)

Abbreviations:

BGI: Building Grid Integration

C: Capacitance

DOE: Department of Energy

EIA: Energy Information Administration

GP: Gaussian Process

HVAC: Heating, Ventilation, and air-conditioning

MOR: Model Order Reduction

MPC: Model Predictive Control

NBI: New Buildings Institute

R: Resistance

RC: Resistance-Capacitance

ZNE: Zero-Net-Energy

Contents

1. Introduction	5
2. RC Modeling Fundamentals.....	7
2.1 RC Model Structure	8
2.1.1 RC Modeling for Envelope	9
2.1.2 RC Modeling for Zone Air	10
2.1.3 RC Modeling for Zone Internal Mass	10
2.1.4 RC Modeling for Internal Heat Gains	11
2.1.5 RC Modeling for Infiltration	11
2.1.6 RC Modeling for a Single Zone	11
2.1.7 RC Modeling for Multi-Zones or Whole Building	13
2.1.8 RC Modeling Diagram Representation	14
2.2 RC Model Creation and Parameter Estimation.....	15
2.2.1 Forward Approach.....	16
2.2.2 Inverse Approach	17
2.2.3 RC Model Order Reduction	18
3. RC Model Applications.....	19
3.1 Heat Dynamics Analysis	21
3.2 Thermal Load Calculation	23
3.3 Building Control and Optimization.....	27
3.4 District/Urban Energy Modeling	31
3.5 Building Grid Integration.....	33
4. RC Modeling Toolbox.....	39
5. Conclusions	40
Acknowledgments	40
References.....	41

1. Introduction

Energy is one of the key drivers of the economy and social development. It impacts multiple business entities, such as energy production and distribution firms, investors, households, and governments. Building energy consumption is greater than industrial and transportation energy consumption in the United States [1,2]. According to the US Energy Information Administration, buildings in the United States consume about 75% of the electricity produced, as well as 55% of the natural gas [2]. Another study showed that buildings account for approximately 40% of total national energy consumption and almost 20% of greenhouse gas emissions; the annual national energy bill totals more than \$380 billion [3].

The US government continuously expands the goals for building energy savings and publishes strict energy standards for new building constructions. For example, according to the California Public Utilities Commission 2008, all new commercial buildings and 50% of the existing buildings must realize net zero energy by 2030, and all newly constructed residential buildings needed to be net zero energy by 2020 [4]. However, reducing the energy consumption is still challenging because of the increasing population and rapid urbanization.

Numerous studies have aimed to reduce building energy consumption with focuses such as advanced controls [5], renewable energy applications [6], and fault modeling and impacts [7,8]. Modeling is the basis in advancing those building technologies. ASHRAE summarized multiple roles of modeling in the building energy field [9], such as thermal load estimation and controls. According to the US Department of Energy [10], modeling provides benefits to numerous fields, including architectural design, HVAC design and operation, building performance rating, and building stock analysis. Other important applications include code compliance, control studies, retrofit designs, and policy analysis. There are more than 100 building energy modeling software available with different domain focuses [11]. Typically, modeling can be classified into three types: white-box, black-box, and grey-box. Each type is summarized as follows.

- (1) The white-box model is fundamentally based on conservation of mass, energy, and momentum. The common approach of making a white-box model is bottom-up (forward): building envelope, occupants, schedules, HVAC systems and parameters, and so on. The modeling is time-consuming, but the simulation results are more accurate than the other models. There are a few prominent software with some white-box models included, such as EnergyPlus [12] and TRNSYS [13].
- (2) The black-box model is a data-driven based modeling approach because of the boosting of machine learning algorithms. The typical requirements for a black-box model are sufficient clean data and appropriate algorithms. It has multiple advantages, including lower engineering cost because it is a data-in-data-out approach; less domain knowledge because it is based on the mapping of input and output data; and greater adaptability because the model will evolve itself with new data. Black-box models also have some drawbacks, including a high demand for data quality: missing, wrong, or biased data lead to low quality models. Also, black-box models lack interpretability and may require intense computation (e.g., GPU), especially with deep learning-based algorithms. For buildings and HVAC systems, black-box models are mainly used for energy estimation or load prediction, which can be subsequently used for controls, energy management, and so on. The typical algorithms include linear regression [14], neural networks [15], and support vector machines [16]. Black-box models are also used for fault detection. Faults are common in buildings and could

cause more than 100% energy consumption [7]. Typical algorithms include Bayesian networks [17,18] and principle component analysis [19].

- (3) The grey-box model has the qualities of both white-box and black-box models. Grey-box models are sometimes called reduced-order models or simplified models. The grey-box model is more interpretable than black-box model. It is more computationally efficient and simpler than the white-box model. The most common grey-box model is a resistance-capacitance (RC) model [20], or thermal-network model.

Because of the emerging building technologies, many challenges must be solved from the modeling perspective. Hong et al. summarized 10 challenging scenarios for the building simulation [21], including urban-scale energy modeling and fault modeling. There are also growing interests in building-grid integration, which demands a faster load calculation at the district or urban level. A review was conducted for simplified models, including the neural network model, linear parametric model, and RC model [22]. However, it did not cover all the essential aspects of grey-box modeling. The thermal-network model (or RC model) became a popular choice as a typical grey-box model. There are two major benefits of RC models:

(1) faster load calculation with major physical dynamics, which is especially helpful for controls and build grid integration; and (2) online controls. RC models can be written into state-space format to be solved by state-space solver for online controls. However, RC models are not widely applied because of various challenges, which can be grouped into five major causes:

- (1) The theoretical limitations and assumptions are unclear.
RC modeling is a good technique for building energy modeling based on the abundance of studies. However, the assumptions and limitations are unclear.
- (2) Grey-box model naming convention and structure are confusing.
For RC models, there are many variants for the order of models, or the naming convention. The definition of RC models is ambiguous to many of the modelers who make grey-box models. For example, some studies use “3R2C” models for wall elements modeling [23,24], and some studies use “3R2C” for zone-level models [25,26]. Some studies found that “3R2C” models would be sufficient for building load estimation [27,28], whereas some studies claim “1R1C” models are sufficient for their studies [29]. Some studies only mention 2nd-order models [30] or 3rd-order models [31,32] for their studied buildings. Furthermore, the diagram of RC models is also unclear: some studies used closed-loop RC representation and others used open-loop representation. In Europe, there is another naming convention, such as the ISO 13790 model [33], ISO 52016 [34]. How to distinguish these modeling structures remains confusing to many researchers.
- (3) Grey-box model creation is vague.
Most studies applied data-driven approach to create RC models [35–37]. However, there is also a forward approach to create RC models. The advantages and drawbacks for the approaches are unclear.
- (4) Suitable applications of grey-box models are unknown.
Many studies used RC models for load or energy estimations [37,38], but other applications are unknown.
- (5) Grey-box models lack a unified software solution for wider adoption.
Few tools are available to create generic grey-box models, and they all have some advantages and disadvantages for wide adoptions and easy operations.

Although a great number of papers study this topic, none of them comprehensively review the topic of grey-box modeling in buildings and address the mentioned challenges in a systematic way. This study aims to provide a systematic overview for grey-box models in the building domain, covering fundamentals, applications, and toolbox. Word cloud statistics are shown in Figure 1 based on the available published studies using grey-box modeling for buildings. As the figure shows, the highest-frequency keywords related to grey-box modeling are closely associated with building, model, thermal, control, and energy, and some others.

Figure 1. Word cloud statistics for grey-box modeling of buildings.

2. RC Modeling Fundamentals

Modeling is the abstraction of the physical buildings and HVAC systems, with some assumptions. Ogunsola and Song [39] identified the common assumptions used in the RC models for buildings, which can be summarized as follows:

There are also some theoretical limitations [39]:

- The varying wind velocity will cause nonuniform convection coefficients for external walls, especially for high-rise buildings. Enhanced external flow dynamics are needed.
- The nonuniform distribution of air can cause nonuniform convective heat transfer, creating issues for windows, or buoyance-driven flows. This defect can be overcome by using computational fluid dynamics.
- RC models do not handle the energy exchange between the return air and supply air well.
- Infiltration is not well modeled.
- The complexity will significantly increase with the number of zones for RC models. This will lead to intense computation burdens. Thus, model order reduction techniques are used.

2.1 RC Model Structure

According to a study by Ljung [40], the grey-box model is defined as a modeling approach in which we can determine multiple parameters of the model from the observed data based on some physical insight available. Ljung further pointed out the grey-box models has two subcategories:

- (1) Physical approach: a grey-box model structure can be established from physical models in which the parameters are estimated from the data. The state-space model with a predetermined order and structure is in this category.
- (2) Semi-physical approach: physical insights are used to guide the data with data-driven models. The cooling capacity performance curves for rooftop unit or chillers fall into this category [12].

Based on the provided definition, the grey-box model is a representation of physical model but ignores minor fidelity and preserves major fidelity. The diagram in Figure 2, demonstrates the concept of RC model creation procedures.

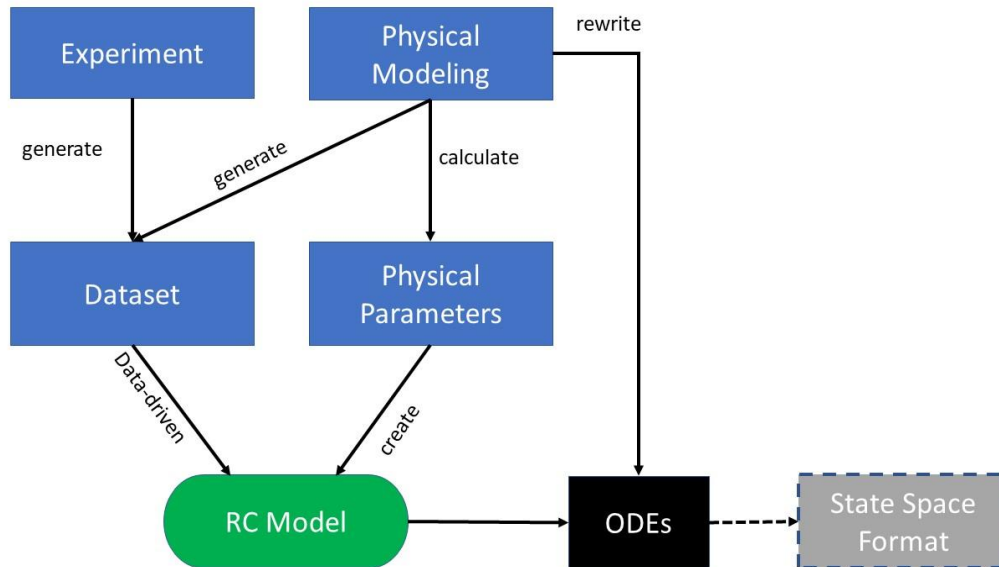


Figure 2. RC model creation procedures.

2.1.1 RC Modeling for Envelope

Based on literature studies, there are numerous representations of RC models. RC models are usually formatted into $xRyC$, in which x is the number of thermal resistances, and y is the number of thermal capacitances. Another naming convention is the z -order model, in which z is the number of thermal capacitances. Even for the same type of RC model, there are multiple naming conventions. To clearly describing RC models, we suggest standardizing the RC models. This section will provide a summary for the RC models.

(1) Window model

For windows, the thermal conduction is dominant. The thermal storage is negligible. The RC model can be treated as 1R0C as shown in Figure 3. With multi-layer windows being adopted, more studies are necessary to confirm the reliability of such modeling.

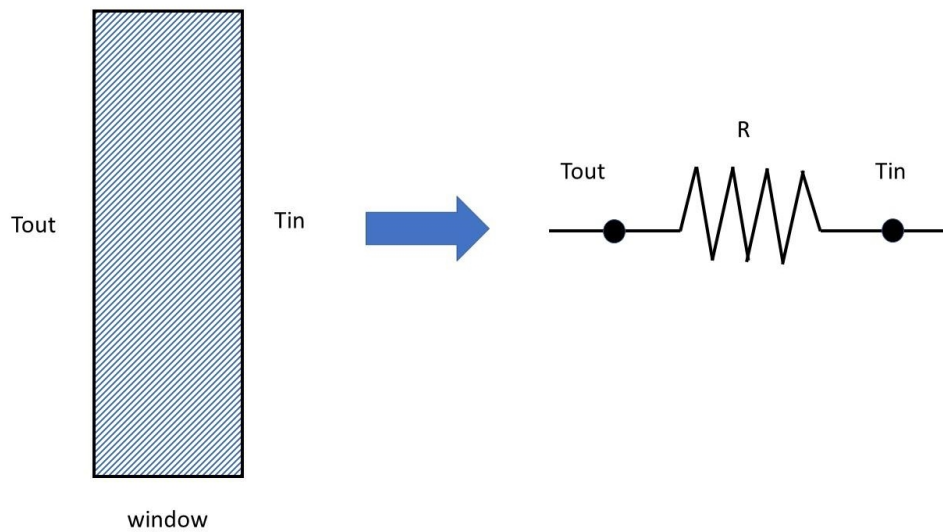


Figure 3. RC model for a window.

(2) Wall model

The walls for building envelopes can be divided into single layers or multiple layers as needed. For walls, there are two heat transfer mechanisms: heat conduction through the wall and heat convection on two sides of the wall surface. For single-layer wall structures, 2R1C models are common (Figure 4). For multi-layer wall structures, the RC models can be shown as 2R1C in (a) and 3R2C in (b) (Figure 4). The 2R1C model has the convection term included into the R terms. The 3R2C has the convection term explicitly. A multi-layer model is usually used for envelope heat dynamic analysis, such as using phase change materials for walls [20].

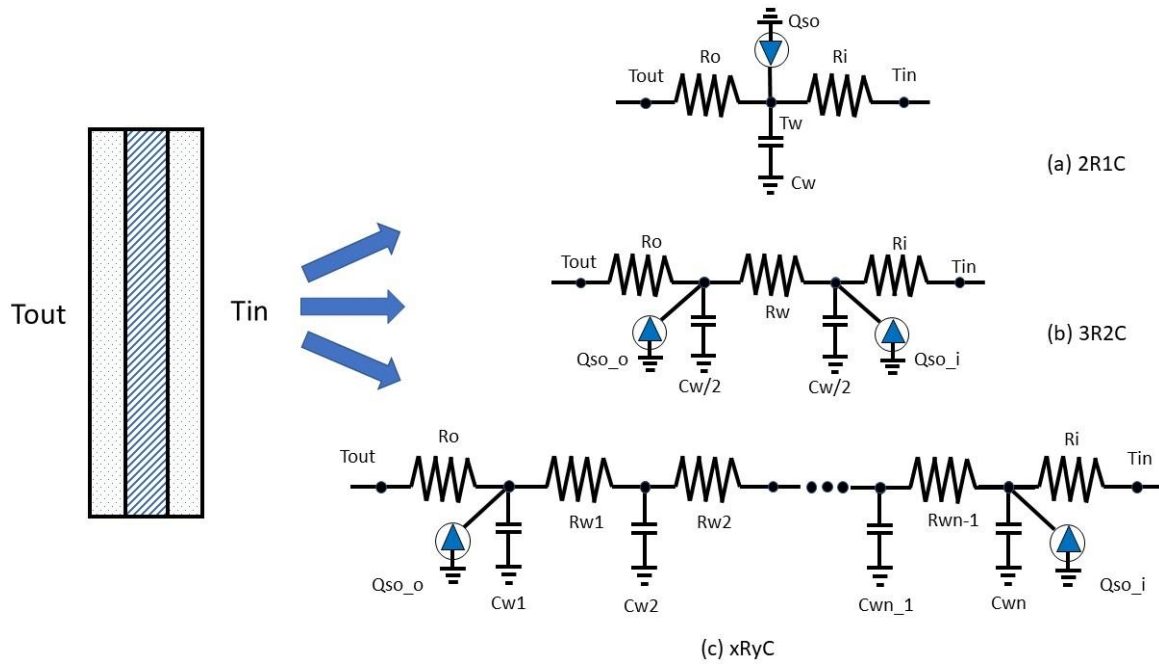


Figure 4. RC model for walls.

Figure 4 shows the standard approach of envelope modeling following the xRyC convention.

(3) Other envelope models

Similar to walls, the roof and floor can be modeled following the same approach. The floor is similar to wall models, except for the ground heat transfer effect. Special boundary conditions are usually required to describe ground heat transfer impact.

2.1.2 RC Modeling for Zone Air

The zone air is usually assumed to be well mixed and uniform. Hence, the zone air is usually modeled as a single capacitor [41].

2.1.3 RC Modeling for Zone Internal Mass

The internal mass is defined as any internal building component influencing the zone loads, such as internal structure, furniture, and carpet. They are optional in RC modeling depending on the required modeling resolution and accuracy. Internal mass can be modeled as a single capacitance connected to the zone air capacitor, or as a pair of thermal capacitances [42]. An example is given in Figure 5.

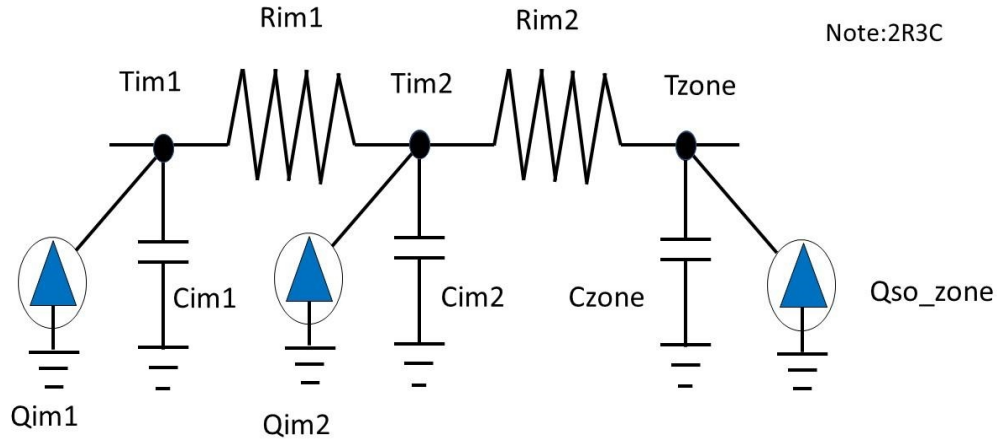


Figure 5. RC modeling for internal mass (recreated based on [42]).

2.1.4 RC Modeling for Internal Heat Gains

The internal heat gains spring from the multiple sources, such as zone occupant heat gains, electrical equipment, and lighting. Typically, they are represented as an ideal, single lumped parameter, similarly with solar radiation to the zone. One example is shown in Figure 7.

2.1.5 RC Modeling for Infiltration

The infiltration is defined as the flow into a zone through openings/cracks in the envelope from outside of the zone. Infiltration has significant impacts on building energy consumption. Excessive infiltration is regarded as a fault due to unexpected openings through envelope [7,8]. To our best knowledge, infiltration is usually assumed to be a disturbance or a lumped parameter.

2.1.6 RC Modeling for a Single Zone

For the zone level, the RC model is established by aggregating walls and windows depending on the investigated building. Figure 6 demonstrates 3R2C-based envelope models for an enclosed space. Usually, a few extra terms (e.g., HVAC cooling/heating, infiltration, occupants, internal mass) need to be added as demonstrated in Figure 6.

(2) Variants for a single-zone model:



(2) Variants for a single-zone model:

12

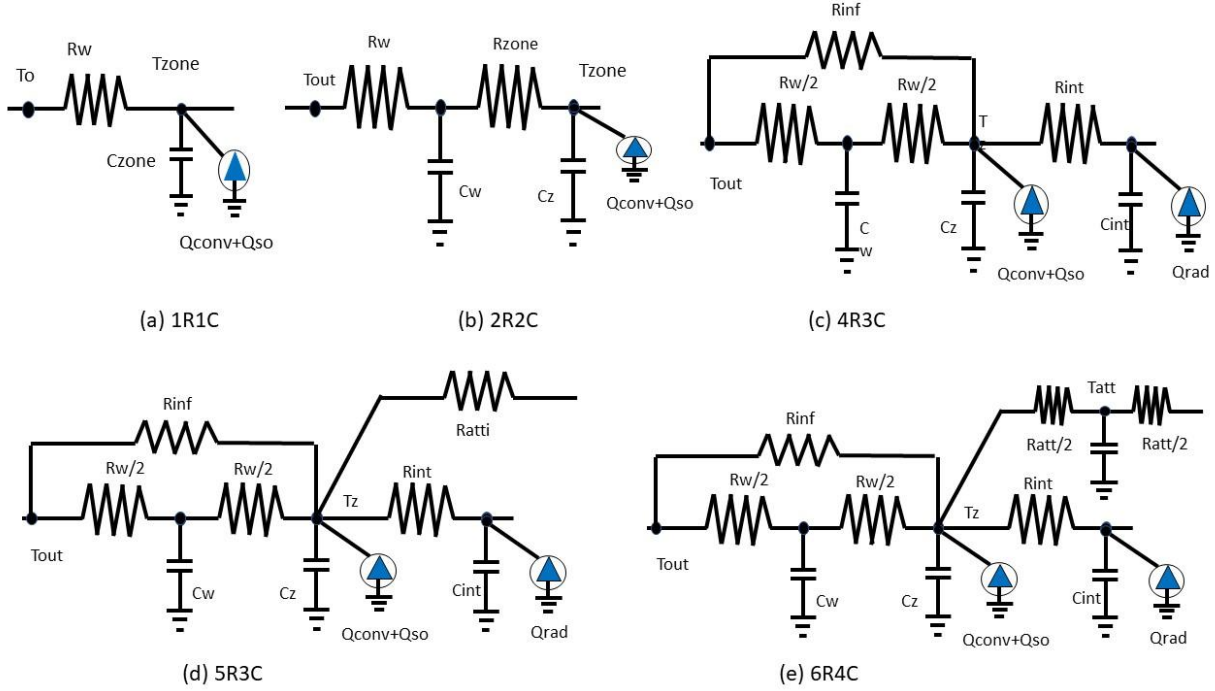


Figure 7. RC zone models (recreated based on [29][36] [43] [44] [45]).

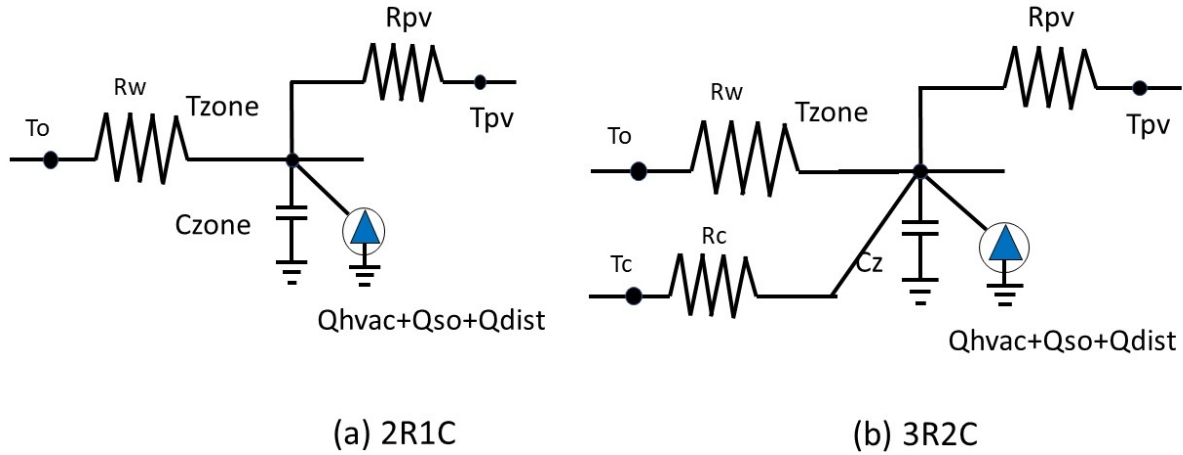


Figure 8. Another Set of RC zone models (recreated based on [29])

2.1.7 RC Modeling for Multi-Zones or Whole Building

Multi-zone models (or the whole building) are based on the repetitions of a single-zone model with shared walls or ceilings. A four-zone building model-based 3R2C was studied by Goyal and Barooah and Deng et al. [27,28]. The diagram following standard modeling conventions for a five-zone model is shown in Figure 9.

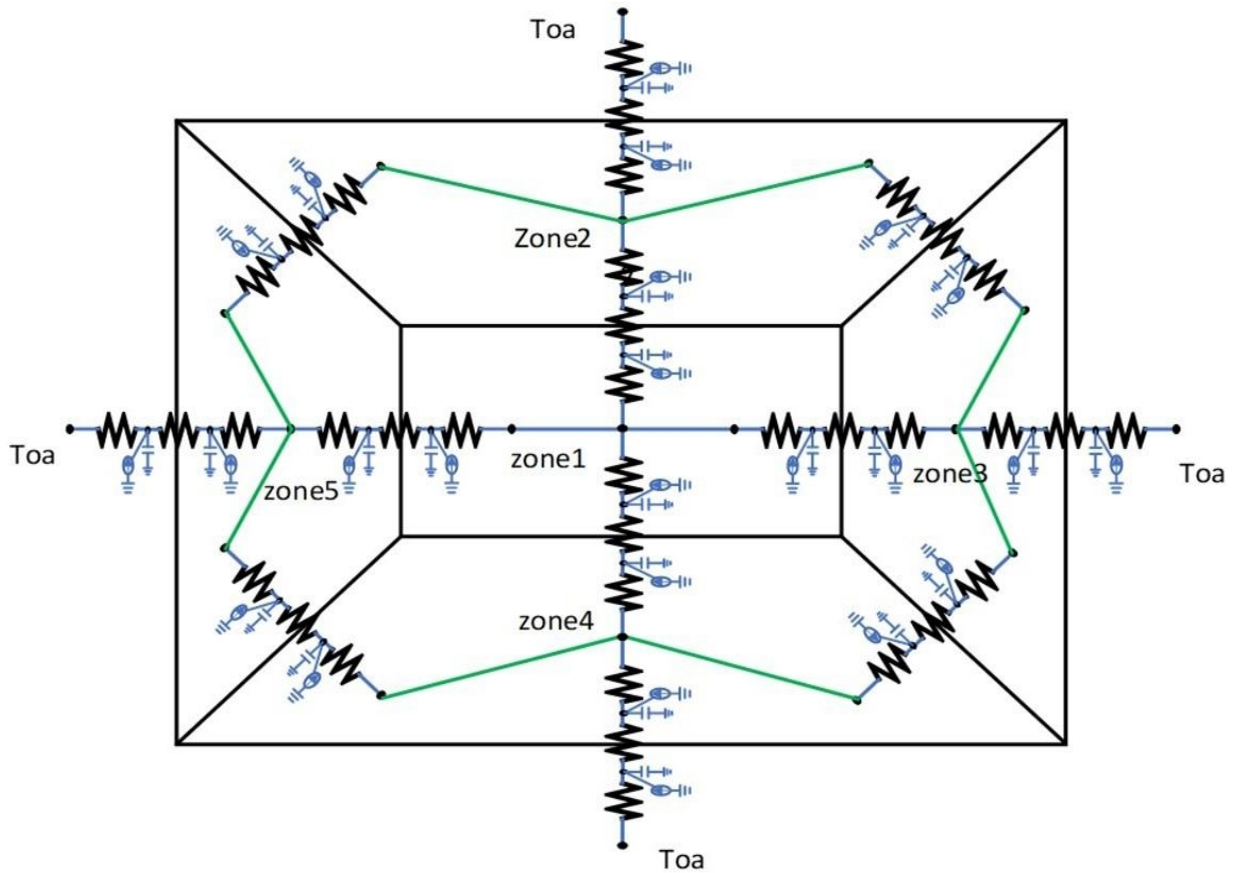


Figure 9. Five zone building model (2D view).

2.1.8 RC Modeling Diagram Representation

Based on the literature, the two typical forms of RC models are open-loop and closed-loop. A diagram is given for closed-loop and open-loop in Figure 10; the top figure (black) is a closed-loop representation for VDI 6007 [46] for a single-zone model, and bottom figure (blue) is a corresponding open-loop representation. The main difference between closed loop and open loop is connecting the shared outdoor node or indoor node to form a closed energy flow like the electric circuit. The temperature difference between two nodes (e.g., indoor and outdoor nodes) is the potential to drive the energy flow.

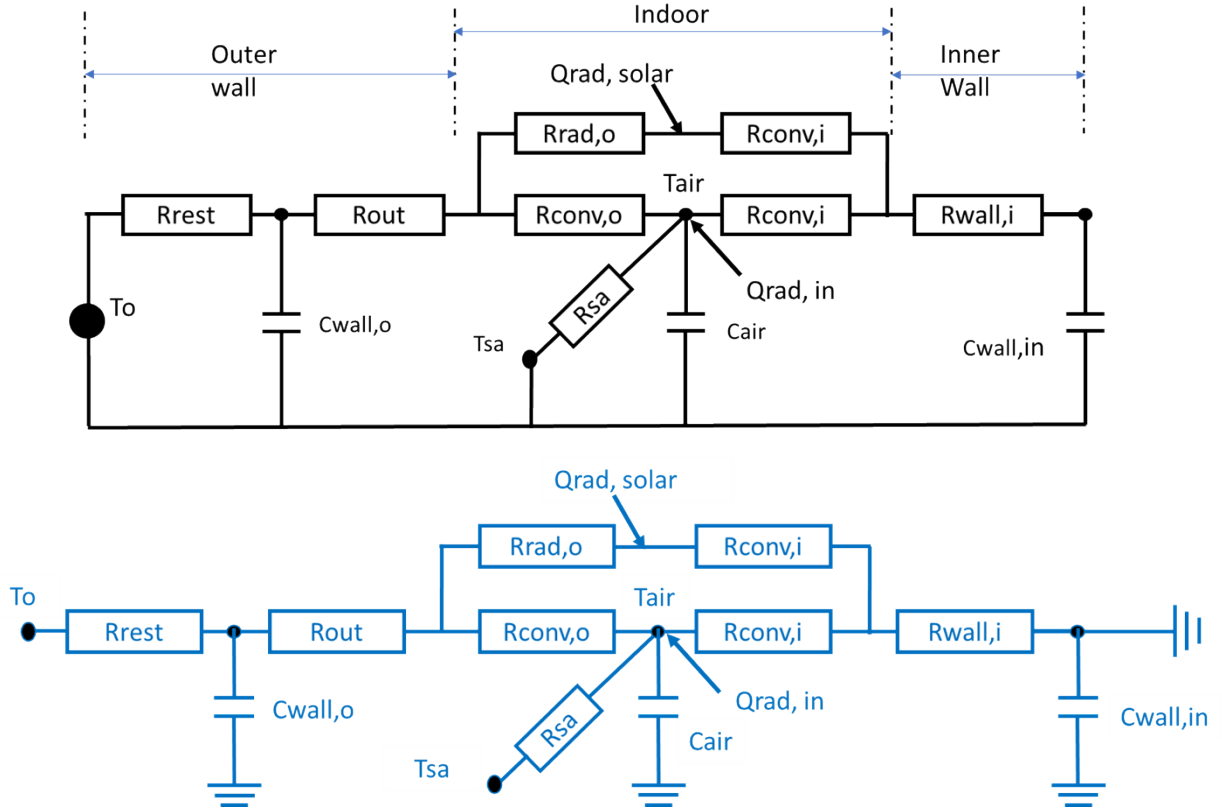


Figure 10. RC room model (top) closed-loop vs (bottom) open-loop forms (recreated based on [46]).

The RC modeling procedure is tedious for multi-zone buildings. The immediate need is to define a clear naming convention for RC models with respect to the envelope, zone, and whole-building level.

2.2 RC Model Creation and Parameter Estimation

A flow chart of establishing RC models for buildings is shown in Figure 2. The most pivotal work for creating RC models is to obtain the R and C values. There are generally two approaches to obtaining the R and C terms [39] as shown in Figure 11. (1) Forward Approach: Direct calculation from physical models achieves the best accuracy. Many building simulation programs can calculate the R and C values (e.g., EnergyPlus [12]). (2) Inverse Approach: Using data-driven algorithms to do curve fitting to get the R and C values. Typical algorithms are least square regression[44,47,48], system identification[39], etc.

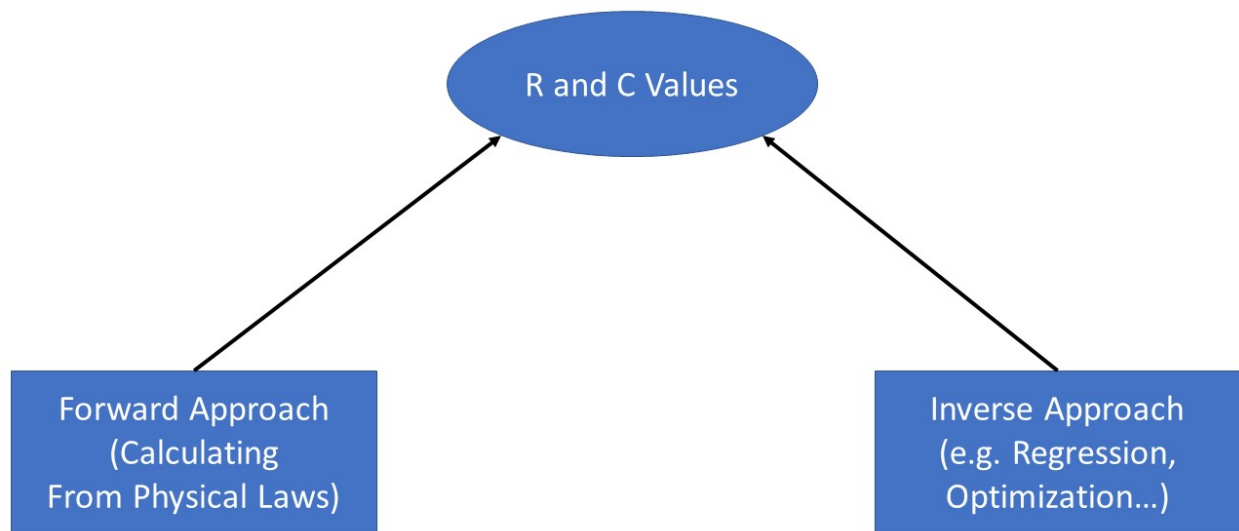


Figure 11. Approaches determining R and C [39].

2.2.1 Forward Approach

The forward approach is shown in Figure 12, which is a bottom-up pattern. Engineers define all envelope parameters (e.g., thickness, thermal conductivity) and then establish the zone level models. The R and C terms are usually available from physical-based modeling software, or experiment data. A couple studies have demonstrated the feasibility of RC modeling using the forward approach[20,49–51]. The two main categories of efforts in this approach are

(1) individual scientific study and (2) software platform development.

(1) Individual scientific study:

A comparison between Gaussian process-based modeling and RC modeling was conducted using a simulated office building. The results demonstrated the RC modeling with the forward approach achieved smaller errors than Gaussian process-based modeling [49]. The pipe-embedded concrete radiant floor was modeled using RC models analytically. The radiant floor had three components: the top cover layer, pipe-embedded concrete slab, and bottom insulation layer. Each layer was modeled as a 2R1C component and coupled together. The numerical results showed that the heat flux prediction for both the top and bottom layers had an error of less than 5.5% compared with the experimental results [20].

(2) Software development:

In 1988, an attempt was made to create a computer program, ELAN, for building modeling by 2nd-order models for multi-zone buildings, focusing on thermal comfort and energy [50]. In the 1990s, a computer program from Tsinghua University was developed to demonstrate the modeling of multi-zone buildings with RC models. The RC models were converted into state-space format for faster convergence [51]. The latest effort is the Modelica-based

modeling platform, TEASER [52]. Another attempt using Modelica was published by Andriamamonjy et al. [53], who proposed an approach to generate grey-box models automatically. This study aimed at facilitating a grey-box implementation procedure by using information from building information modeling to generate different model structures that reflect building-specific geometry and design.

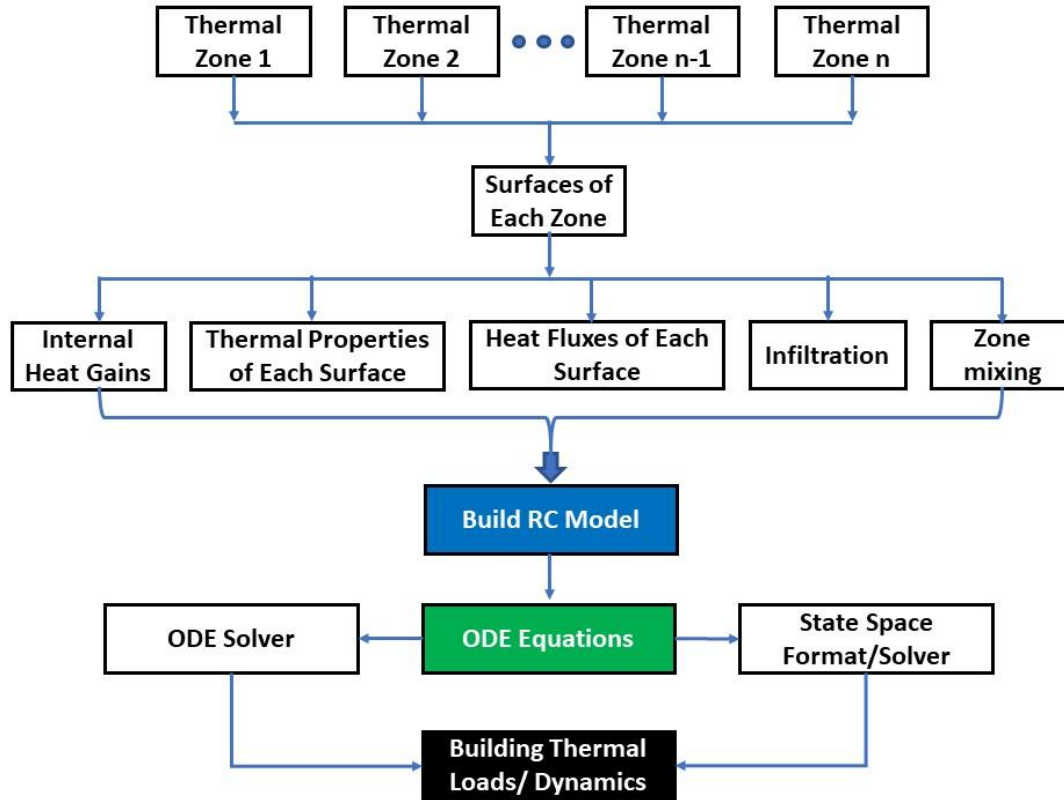


Figure 12. Flow chart of building grey-box model (forward approach).

2.2.2 Inverse Approach

The inverse approach uses the data-driven technique (e.g., regression) to obtain the R and C terms. The majority of published studies applied this approach. A flow chart for this approach is shown in Figure 13. The cleaned data set and RC model structures are prerequisites for model establishment. The system identification is usually applied to derive the parameters in the RC models.

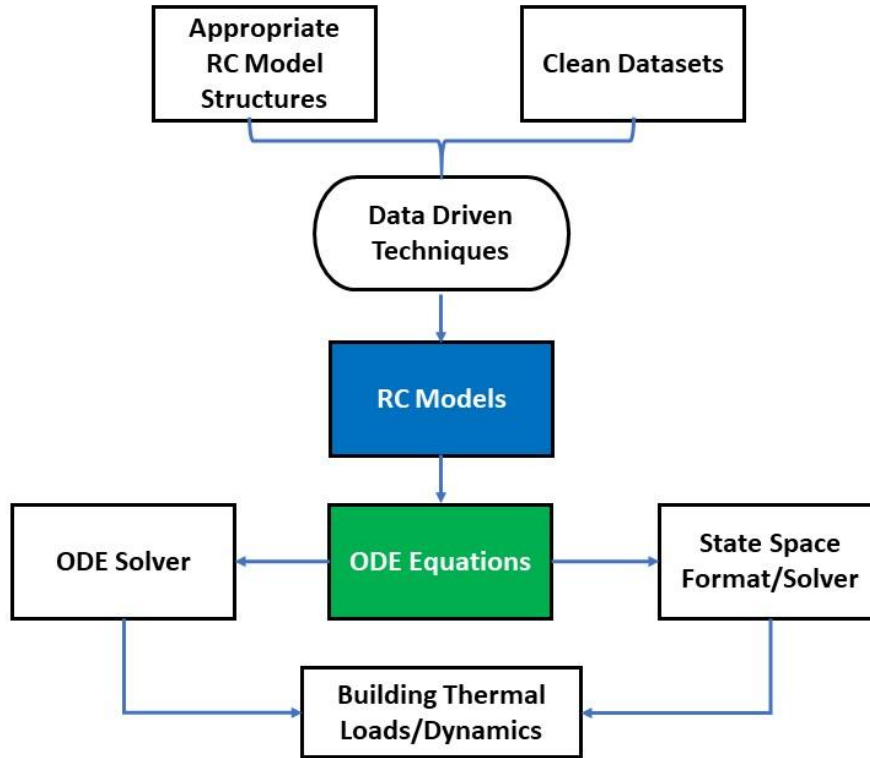


Figure 13. Flow chart of grey-box model (inverse approach).

Many methods conduct identifications [54–56]. Okuyama and Onishi applied the system identification to estimate the parameters of RC building models for a two-story building with an area of 200 m². They predicted the heat transfer parameters, such as coefficient of heat loss, solar heat gain, effective thermal capacity, and infiltration rate. The modeling results had good agreement with physical-based software simulations [57]. A new system identification approach was proposed to estimate the parameters of RC building models with better accuracy for a multi-zone lab room under unmeasured disturbances [58].

2.2.3 RC Model Order Reduction

Because RC models are mathematically a set of ordinary differential equations (ODEs). For ODEs, the number of orders is equivalent to the number of C terms. Because of the computation burdens, model order reduction methods typically need to be applied to discard the less impactful terms from full-order models. A series of order reduction methods is available, which can be found from Antoulas et al. [59], with the theory presented by Antoulas [60]. Here, only the model reduction methods related to RC modeling are reviewed. The model order reduction methods provide insight regarding what model order would be sufficient to reduce the computation burdens.

Generally, model order reduction can be divided into three categories [61]: polynomial reduction methods, optimization approaches, and state-space transformation methods. The polynomial reduction methods are based on the frequency domain [62]. The optimization

approach is based on the time domain, which uses parameter estimation to build the reduced-order models. The state-space approach used the stable state coordinates. A detailed example using of the state-space approach can be found from Kim and Braun [63]. Another example of the state-space transformation approach is available using a linear model reduction technique mathematically for building energy modeling with state-space formulations [64]. The study showed the feasibility of theoretical considerations and practical implementations for the heat flux study on two sides of the wall. The results showed that heat flux was within 3% accuracy for the 1st-order model, and within 0.5% accuracy for the 2nd-order model, compared with the baseline case. However, the authors did not present the zone air temperature accuracy from those reduced-order models.

Deng et al. [65] proposed a new model order reduction method for a four-zone building from a full-order building model to a 3R2C model. A further demonstration of their proposed model order reduction was shown by Goyal and Barooah [27]. Their method works by aggregating the nodes into super-nodes to reduce the model orders. In another study, Gouda et al. compared the 2nd order and 20th order building models, and their results showed that the 2nd order is sufficient in comparing with the baseline model for high- and low-thermal capacity buildings [61]. Menezes et al. studied the heat transfer of building envelopes [66] for a single-zone building and discussed the coupling strategies with other models. They concluded that the proposed model reduction reduced the computation burden while maintaining an acceptable accuracy. Another study [67] used state-space modeling techniques to identify the parameters with the inverse modeling approach for a large residential building complex. The authors investigated the performance of multiple RC models with different orders. They concluded that the state-space technique was much faster than other techniques: it only took 4.5 s to finish simulations of 100 years on a i5-processor computer.

Overall, a standardized model order reduction tool for building thermal modeling is still desired, which requires more knowledge from applied mathematics or numerical mathematics.

3. RC Model Applications

Because of the merits of RC models (e.g., simple, robust, fast, computationally efficient [68]), they have been widely used in multiple applications in the building energy domain. This section summarizes multiple major applications of grey-box models as shown in Figure 14. The load estimation or energy prediction is essential for other major applications (e.g., control and optimization, building-grid integration). Based on the reviewed cases, a simple statistic is given in Figure 15: (a) Thermal load calculations dominate other applications; (b) Inverse approach has more cases than forward approaches; (c) Sing-zone thermal load calculation has more studies than multi-zone scenarios; (d) Second-order grey-box modeling is the top pick, comparing with other orders.

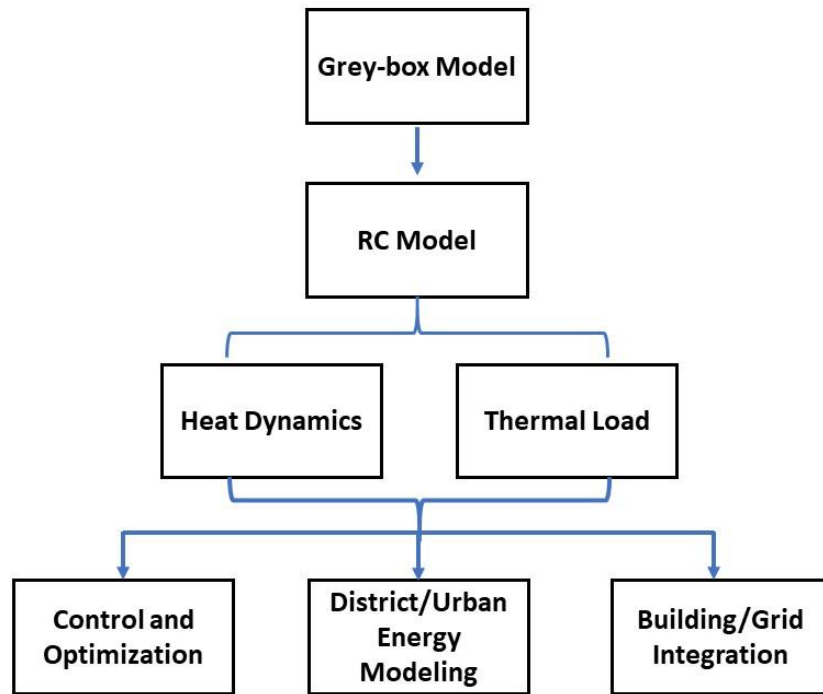


Figure 14. RC model applications in buildings.

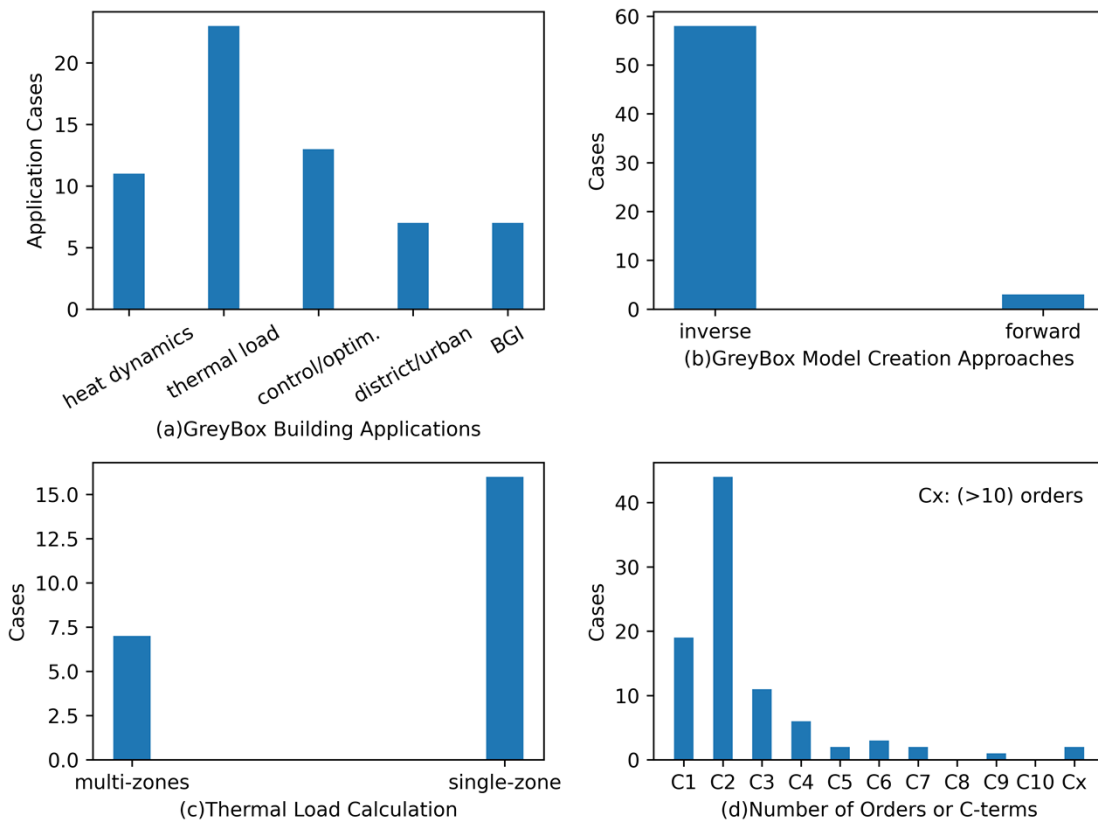


Figure 15 Grey-box Modeling Statistics

3.1 Heat Dynamics Analysis

The heat dynamic analysis is focusing on investigating heat transfer properties of building envelopes, for example phase-change-material integrated walls, or radiant floors. Those studies are usually experimental investigations that employ RC modeling as an auxiliary. A summary is shown in Table 1. A timeline in Figure 16 shows growing interests of RC models for building heat dynamics calculation, especially innovative building wall materials.

Table 1. Heat dynamics analysis.

RC model	RC parameter	Variables studied	Envelope	Experiment or modeling	Reference
6th order	Identification	Zone temperature, heat flux	Radiant cooling/underfloor ventilation	Both	[47]
1st order	Identification	Surface temperature, heat flux	General walls	Modeling	[69]
3R2C	Identification	Heat transfer coefficient	Solid walls/cavity walls	Modeling	[23]
4R2C	Identification	Surface temperature, heat flux	Walls with phase change materials	Modeling	[70]
1R1C	Identification	Zone temperature	General walls	Modeling	[71]
4th order	Average value	Thermal conductivity, heat flux	Photovoltaic thermoelectric wall systems	Both	[72]
9th order	Optimization	Heat flux	General walls, including concrete slabs	Both	[73]
2R1C	Identification	Heat flux	Pipe-embedded concrete radiant floors	Both	[20]
4R3C	Identification	Zone temperature	General walls for multi-zone	Modeling	[74]
5R1C	Identification	Zone temperature	General walls for single building	Modeling	[75]
2R1C	Identification	Zone temperature	General walls	Modeling	[76]

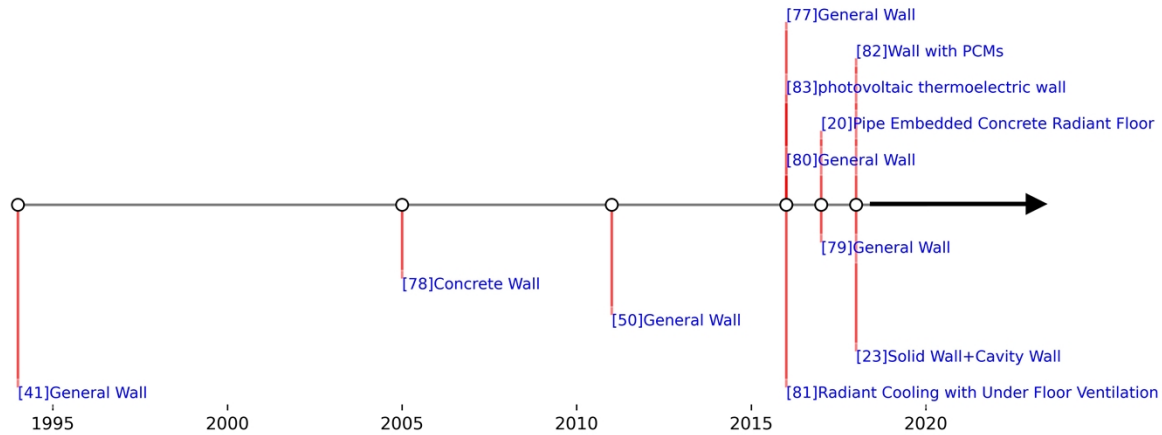


Figure 16. Timeline of RC model for heat dynamic analysis. PCMs = phase change materials.

This group of applications can be categorized into (1) general wall heat dynamics analysis (2) innovative wall heat dynamics.

(1) General wall heat dynamics analysis

A new analytical solution was proposed to change the characteristic terms in RC models for the first-order model of a building [69]. The proposed solutions improved the accuracy of RC models through a self-adjusting manner. The modeling results showed good accuracy in estimating surface temperature, zone temperature, and heat flux. Park et al. identified the thermal resistor and thermal capacitor for the 1R11C model from the experimental results in a test room based on the heat transfer equations. They obtained a satisfactory model representation of the building (1R1C) for the indoor air temperature estimation [71]. Weber and Johannesson compared the star-type RC model and triangle-type RC model for building walls [73]. The star type can be only used for concrete slabs. Their results showed that the triangle RC models had wider applications, such as radiant heating, radiant cooling, and concrete core conditioning systems. A 4R3C model was applied in a building to study the impact of adjacent wall on zone thermal loads. The TRNSYS modeling results were used to identify the parameters of RC models. The modeling results agreed well with the TRNSYS simulation of zone thermal loads [74]. A 5R1C model was developed according to EN ISO 13790 for a double skin test cell. The zone temperature calculations showed good agreement with the measurement data [75].

(2) Innovative wall heat dynamics

Innovative walls use composite materials to change the original thermal properties. A study was conducted in a single-zone room with radiant cooling system integrated with underfloor ventilation by using RC models. The least squares method was used to identify parameters such as thermal resistance, thermal capacitance, and heat transfer coefficients. The modeling results had small errors compared with the experimental results: differences of less than 0.5 K for zone temperature and less than 7.7% for heat flux [47]. The error analysis was conducted for a single-zone building to calculate the U value based on RC models for a solid wall and a cavity wall. The results showed that the 3R2C model had better accuracy [23]. The variable-resistance-variable-capacitance model was proposed for a wall with phase change materials. The results showed that 4R2C model was more favorable with smaller

errors compared with the experiment data [70]. The surface temperature difference was 0.42°C, and surface heat flux was about 15% compared with measurement data. Luo et al. studied the photovoltaic thermoelectric wall system as a new type of energy saving envelope using the state-space format. They built the RC-based thermal dynamic models for this wall and photovoltaic panels and demonstrated 70% daily heating energy saving compared with traditional walls [72]. The pipe-embedded concrete radiant floor was analytically modeled using RC models. The radiant floor had three components: the top cover layer, pipe-embedded concrete slab, and bottom insulation layer. Each layer was modeled using 2R1C. The numerical results showed that the heat flux prediction for the top and bottom layers had an errors of less than 5.5% compared with the experimental results [20]. Mathews et al. proposed a fast building design approach, which applied the 2R1C wall element model for buildings. They accomplished 70 case studies from 32 buildings with diverse thermal characteristics, including experimental buildings, residential buildings, shops, storerooms, schools, high-rise office buildings, factories, and church buildings [76]. The simulation results showed that such a model approach was acceptable with a zone temperature estimation error of less than 2°C compared with the reference data.

3.2 Thermal Load Calculation

There are more studies of using RC models to calculate thermal loads than other applications. The orders of RC models vary widely across different investigations. A summary is shown in Table 2. A timeline is shown in Figure 17, which demonstrates increasing application of RC models for thermal load estimation in recent years.

Table 2. Thermal load calculation.

RC model	RC parameter	Variables studied	Multi-zone or single-zone	Experiment or modeling	Reference
5R1C	Average value	Heating/cooling load hourly	Single	Modeling	[77]
2R2C, 3R2C, 4R2C, 5R2C	Identification	Thermal load	Single	Modeling	[78]
5R1C, 7R2C	Average value	Thermal load	Single	Modeling	[79]
5R1C	Average value	Thermal load	Single	Modeling	[80]
5R1C	Identification	Thermal load	Multi-zone	Modeling	[38]
3rd order	Identification	Thermal load and zone temperature	Single	Modeling	[30]
3R2C, 2R2C	Identification	Thermal load	Single	Modeling	[81]
1R1C, 2R2C, 3R3C	Identification	Heat flux and zone temperature	Single	Modeling	[82]
2nd, 3rd, 4th, 5th, 6th, 7th order	Identification	Heat capacity, thermal Resistance, UA-value	Multi-zone	Both	[31]

1R1C, 3R2C, 4R2C, 8R3C	Optimization	Thermal load and zone temperature	Single	Modeling	[83]
3R3C, 2R1C	Optimization	Energy consumption	Multi-zone	Modeling	[84]
3rd order	Optimization	Energy consumption	Multi-zone	Modeling	[32]
2R2C	Identification	Energy consumption and zone temperature	Single	Modeling	[85]
3R2C, 2R2C	Identification	Thermal load	Single	Modeling	[25]
3R2C	Direct calculation	Thermal load and zone temperature	Multi-zone	Modeling	[24]
3R2C	Direct calculation	Thermal load and zone temperature	Single	Modeling	[26]
3R2C	Identification	Thermal load and zone temperature	Multi-zone	Modeling	[86]
3R2C	Identification	Thermal load and zone temperature	Single	Modeling	[87]
3R2C, 2R2C	Identification	Thermal load	Single	Modeling	[42]
40th, 14th order	Identification	Thermal load	Single	Modeling	[61]
2R2C	Identification	Thermal load	Single	Modeling	[44]
7th order	Direct calculation	Thermal load and zone temperature	Multi-zone	Modeling	[88]
3R2C	Optimization	Thermal load	Single	Modeling	[89]

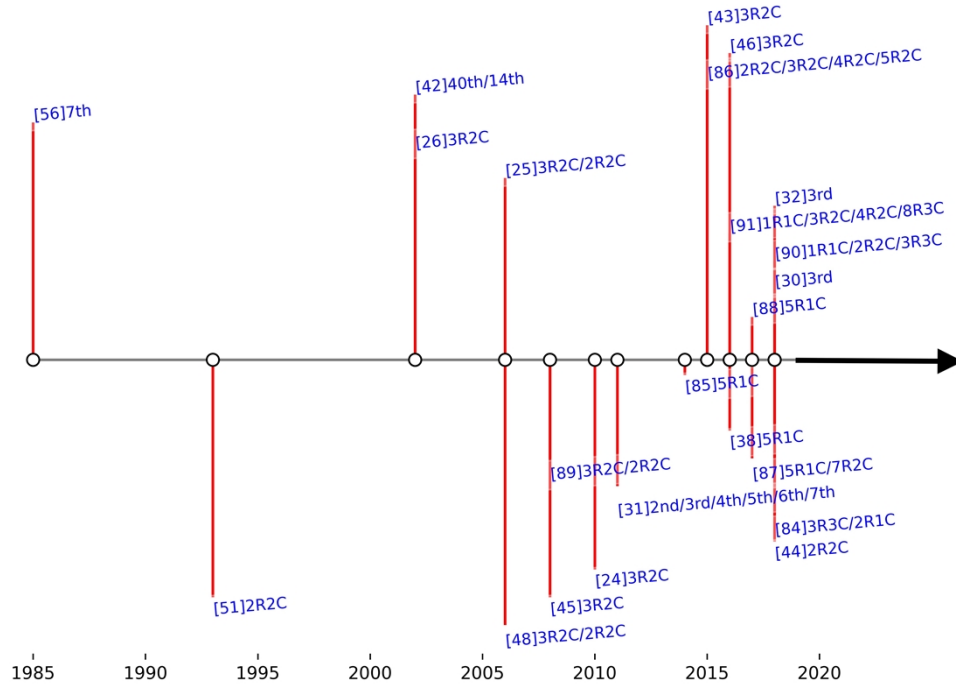


Figure 17. Timeline of thermal load by grey-box modeling.

This group of applications can be categorized into (1) single-zone building and (2) multi-zone building.

(1) Single-zone building

The 5R1C model according to EN ISO 13790 was used to model a residential building. The hourly cooling and heating load agreed well with EnergyPlus simulation results [77]. Analytical solutions were proposed for simplified RC building models with a set of different orders: 5R2C, 4R2C, 2R2C, and 1R2C [78]. The results showed that the proposed solutions had good agreement with the measurement data and EnergyPlus results. A comparison was performed between 5R1C and 7R2C models to a residential building based on ISO13790 and German Guideline VDI 6007, respectively. The results showed that the 7R2C model had better accuracy and was acceptable for heating and cooling load prediction [79]. A 5R1C model with time varying parameters was developed in Simulink according to EN ISO 13790 for a single-zone building. Results showed that the heating and cooling loads agreed well with EnergyPlus results [80]. However, another study concluded that the 5R1C model based on EN ISO 13790 was not accurate enough for cooling and heating load prediction compared with results from TRNSYS [38].

A hybrid modeling effort was implemented for a single-zone building using the lumped RC building model and Gaussian process. The results show acceptable accuracy compared with TRNSYS regarding zone air temperature and energy consumption [30]. Gouda et al. compared the building model reductions by investigating the order of RC models from 1st to 20th order to the building wall elements for a single-zone room through optimizations. The zone air temperature results showed that the 3R2C model had the same accuracy with the 20th order with much less computational effort [61]. Coffman and Barooah identified the

linear models for a one-zone auditorium space with unmeasured disturbances (occupant-induced loads) from the experiment data. They applied the 2R2C model for the building [85]. The zone air temperature was estimated under dynamic occupant scenarios. However, they did not explore the capability of applying this modeling for nonlinear building models and multi-zone buildings.

Dewson et al. applied least squares parameter identification to generate the air temperature prediction model using time series measurements from a test cell. The model (3R2C) results agreed well with the reference data [44]. Braun and Chaturvedi applied the field test data to train the 3R2C simplified building model and obtain the thermal parameters of the multi-zone building. Their load calculation results show the simplified building model agree well with field test [26]. Wang and Xu built a simplified building model based on mixed RC models (3R2C for the external walls and roof, 2R2C for the internal walls) and applied the genetic algorithm to identify the thermal resistances and thermal capacitors [25] for a high-rise building. The simplified models predicted the cooling load and indoor air temperature with good accuracy compared with the field test data. Zhou et al. applied the same RC models [25] with weather prediction to predict the thermal load [81] for energy-efficient control. Kim and Braun proposed a general approach to create RC models for large multi-zone buildings [63]. They also demonstrated the satisfactory modeling accuracy comparing TRNSYS with RC models.

Danza et al. applied a 3R2C model in control to improve the building performance [87]. Their results agreed well with the measurement data, including the zone temperature, hourly heating flux, and cumulative daily energy needs. Rouchier et al. proposed the model calibration for a one-zone residential building [82] in zone temperature prediction using RC models 1R1C, 2R2C, and 3R3C. Their calibration results showed that higher-order models have better accuracy. They also demonstrated the 1st-order model to be biased. Wang and Xu used using the genetic algorithm to identify the internal thermal mass of single-zone building model, in which they used 3R2C and 2R2C models to represent envelopes and internal mass, respectively [42]. Their prediction for thermal load and zone air temperature agreed well with the reference data.

Harb et al. studied various grey-box models for building indoor air temperature forecasting [83], including 1R1C, 3R2C, 4R2C, and 8R3C. Their results showed that the 4R2C model had the best model accuracy (0.3 K) with measurement data for two office buildings and one residential building. Harish and Kumar identified the parameters for a 3R2C-based building model for building load estimation through optimization techniques [89].

(2) Multi-zone building

Bacher and Madsen [31] proposed a procedure to identify the building thermal dynamics for a multi-zone lab room using the RC models. The modeling results agreed well with the experiment data. Another study [84] applied the 3R3C model for the building interior and 2R1C model for the building foundation to study the building energy consumption compared with the EnergyPlus model. They demonstrated that the proper thermostat setback could help save 104 MWh of potential energy for school building mixing with office, classrooms, and labs. A third study [32] applied RC models in Modelica for energy auditing. The results

showed that the reduced-order modeling results had acceptable accuracy compared with both detailed physical modeling results and the measurement data. O'Neill et al. applied the 3R2C model to estimate the thermal load for a multi-zone building through the Kalman-filter estimation, which was validated by the measurement data [24]. Lee and Braun applied the 3R2C-based thermal network model to the Energy Resource Station building (a multi-zone building) for the Iowa Energy Center [86]. They tested the demand-limiting control with an optimal setpoint that minimized the peak cooling demand during a demand-limiting period. The results show the cooling load prediction was within 5% accuracy with 30% reduction in peak cooling load compared with the conventional control.

3.3 Building Control and Optimization

According to ASHRAE [9], building load estimation is the basis for controls, including secondary controls and primary controls (Figure 18).

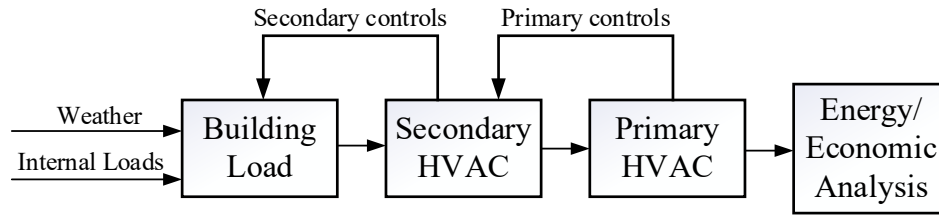


Figure 18. Roles of building load.

Typically, for control and optimization, two targets are sought from RC models: the zone temperature estimation and thermal load calculation. The controls can be classified into the local control and supervisory control (e.g., model predictive control [MPC]). RC models are generally used for supervisory control, especially the MPC controls. A summary of RC models for controls is shown in Table 3. A timeline is shown in Figure 19, in which the RC models have been applied consistently for control and optimization.

Table 3. Building control and optimization.

RC model	RC parameter estimation method	Control	Control variables	Modeling or experiment	Reference
3R3C, 2R1C	Identification	Thermostat setback	Zone temperature	Modeling	[84]
4th order	Identification	MPC	Energy consumption	Modeling	[90]
3R2C	Identification	MPC	Energy consumption	Modeling	[91]
1st order	Identification	MPC	Energy consumption	Both	[92]
2nd order	Identification	MPC and heuristic control	Energy consumption	Both	[93]
2R2C	Identification	MPC with feed-forward control	Energy consumption	Both	[94]

2R2C	Identification	Model-based control	Energy consumption and zone temperature	Both	[95]
3R2C	Identification	MPC	Energy consumption and zone temperature	Modeling	[96]
3R2C	Identification	MPC	Energy consumption and zone temperature	Both	[97]
5R4C	Identification	MPC	Energy consumption and zone temperature	Both	[98]
2nd order	Identification	MPC	Energy consumption and zone temperature	Both	[48]
2nd order	Identification	MPC	Energy consumption and zone temperature	Both	[99]
1R1C, 2R1C, 3R2C, 6R4C	Identification	MPC	Zone temperature	Both	[29]

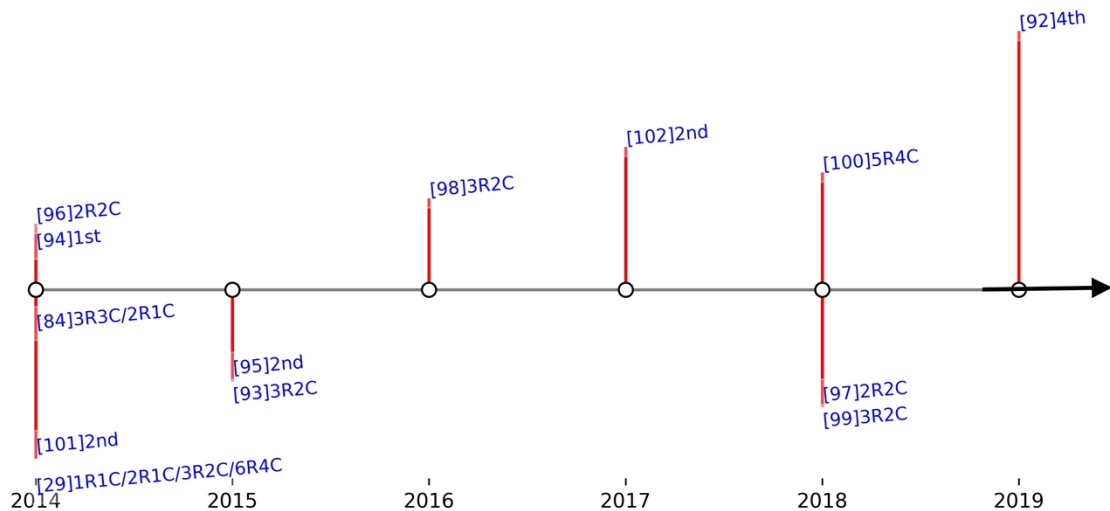


Figure 19. Timeline of control and optimization by grey-box modeling.

This group of applications can be categorized into (1) other (non-MPC) supervisory controls and (2) MPC controls.

(1) Other (non-MPC) supervisory controls

Rosin [84] applied a combination of RC models for building energy consumption: an 3R3C model for the building interior and an 2R1C model for the building foundation. This model

was coupled with a simplified HVAC model to predict total annual energy consumption by indoor zone temperature. Rosin demonstrated that the proper thermostat setback could potentially save 104,000 kWh for a 50,000 ft² school building during unoccupied hours. In a study from Shan et al. [95], a simplified building RC model and a black-box model were combined by a simple data fusion algorithm for easy implementation and accurate prediction. The RC model predicted the room temperature during the period when limiting control was activated. Based on that prediction, the time to activate the limiting control could be determined. The feasibility of such energy recovery was validated on-site in a super-high-rise commercial building in Hong Kong, and the proposed method was validated on a dynamic simulation platform built using the same building. In the two on-site validations, the energy savings during the energy recovering period were 85.8% and 80.1%, respectively.

(2) MPC controls

Li et al. [90] developed a 4th-order RC network model for an office building that integrated photovoltaic-thermal collectors, radiant floor heating, and a thermal energy storage tank to enable optimal collection and use of solar energy. A detailed TRNSYS building model was considered as a ground-truth building and used to identify the parameters of low-order linear time-invariant state-space models. Based on the state-space model of the office building, an MPC algorithm for the optimal set-point trajectory of the thermal energy storage tank was developed and demonstrated to save 34.5% of the electrical energy consumption of the heat pump.

Hu and Karava [91] presented a control-oriented modeling approach for multi-zone buildings with mixed-mode cooling based on the linear state-space representation (RC model structure) with varying coefficient matrices. Key features were the time-variant thermal resistances associated with the heat extraction due to airflow that is calculated using an airflow network model. This approach was validated with the experimental data collected in a two-zone test building under four operation modes.

Huang et al. [92] proposed a new type of MPC scheme based on neural network feedback linearization to achieve energy savings for a commercial airport building. In the proposed control framework, the thermal dynamics of the building were modeled by a linearized RC model, and the uncertainties associated with the HVAC process were handled separately by an inverse neural network model. This approach was tested at the hall of the T-1 building of the Adelaide Airport through simulations and a field experiment. The merits of the proposed method compared with the existing control method were analyzed from energy-saving and cost-saving perspectives. The results showed that the proposed scheme performed reasonably well and achieved a considerable amount of savings without lessening thermal comfort.

A recent study applied the thermal networks building models in MPC to study the control strategies of a multi-zone building for mixed-mode cooling: open-window, fan-assist, and night-cooling [93]. The results showed that 83% of energy savings were achieved by the proposed controls.

Zhuang et al. [94] proposed a new modeling method for a medium-sized commercial building. A MPC strategy with a feed-forward control structure was developed to increase

the supply chilled water temperature. A 1R1C structure was applied to predict indoor air temperature from outdoor air temperature. Field testing in a medium-sized commercial building was implemented to evaluate the MPC strategy. The results showed that a considerable amount of energy saving was achieved through the proposed MPC.

In the study from Fux et al. [29], several lumped-parameter thermal models of a passive house with an integrated photovoltaic system were compared to evaluate the model complexity necessary to capture the basic thermal behavior of the entire building. In order to reduce implementation costs, the state and parameters of the chosen 1R1C model are estimated online with an extended Kalman filter. Based on the predicted system behavior, an MPC controller generated the optimal inputs (or control commands) for the HVAC system. The self-adapting model guaranteed an energy-optimal performance of the building services during various operating conditions, and the comfort of the occupants was maintained at the highest possible level. The results show that the extended Kalman filter yielded a robust convergence in parameter estimation after approximately three weeks. Also, the adapted model generated a prediction of the heat demand for several days.

Salakij et al. [96] developed a physics-based building hygrothermal model and model-based predictive control method to perform the optimal control in building energy management. To obtain a relatively accurate but not overcomplicated model, a building energy analysis model was used based on a lumped-capacitance 3R2C model. The developed building energy analysis model was verified with an analytical solution and compared with simulation results from EnergyPlus. Results showed that the model-based predictive control saved thermal energy consumption up to nearly 43% compared with a traditional temperature control with a constant setpoint.

Yang et al. [97] proposed a method to develop an integrated state-space model for indoor air temperature, radiant temperature, humidity, and predicted mean vote index suitable for fast, real-time, multi-objective optimization. The RC model was adopted to represent the integrated heat and moisture dynamic in the building to reduce the number of states. Using the developed model, a multi-objective MPC controller was proposed and its performance was evaluated through a case study on the BCA SkyLab test bed facility in Singapore. Compared with the conventional on/off control, the MPC controller achieved up to 19.4% energy savings while keeping the predicted mean vote index within an acceptable comfort range.

Viot et al. [98] presented a method for model selection (type, level of complexity) and implemented the developed model in an MPC controller to dictate the control of a long time response floor heating system on a real building. Short measurement periods were carried out to identify the model parameters that minimize the gap between the model output and measurement. Gray-box models (RC models) based on electrical analogy and state-space representation were proposed. They were constructed from physical knowledge and then identified by choosing the most appropriate measurement series. A sensitivity analysis method (Morris) was also used to improve the quality of the identified model, which satisfies control criteria with two specific validation measurement series.

Hazyuk et al. [48] investigated model predictive controller performances and compared them with those obtained by proportional-integral-derivative control and scheduled start proportional-integral-derivative controllers. The model predictive controller required a model of the system (a low-order model is usually preferred). The employed modeling approach obtained a low-order building model in two steps. First, an RC model structure (or a lumped-capacity model structure) was obtained for a thermal circuit representation of the building. Then the values of model parameters were estimated by the least squares identification method using input/output data sets obtained by the simulation of the detailed building model.

Zong et al. [99] presented an economic MPC strategy for energy management in smart buildings, which can act as active users interacting with smart energy systems. In the case study, they built an RC model for a residential building called PowerFlexHouse3. The economic MPC strategy based on the RC model of the residential building outperformed the traditional proportional-integral-derivative schemes for energy saving and load shifting.

3.4 District/Urban Energy Modeling

Because of the capability of fast load calculation of RC models, district/urban energy modeling has emerged in recent years. A summary is shown in Table 4. A timeline of RC modeling for district/urban energy modeling is shown in Figure 20. RC modeling has gradually gained more popularity for district/urban level energy calculations.

Table 4. District/urban energy modeling.

RC model	RC parameter estimation method	Modeling scope	Predicted variables	Note	Reference
3R2C	Identification	Urban	Energy consumption		[68]
5R1C, 2nd order	Identification	District	Energy consumption	ISO13790, VDI6007-1	[46]
1st, 2nd, 6th order	Identification	Urban	Energy consumption	Modelica	[100]
3R2C	Identification	District	Heating energy	Modelica	[101]
2nd order	Identification	District	Energy consumption	Modelica	[37]
3rd order	Identification	Urban	Energy consumption		[102]
2nd order	Identification	Urban	Energy consumption		[103]

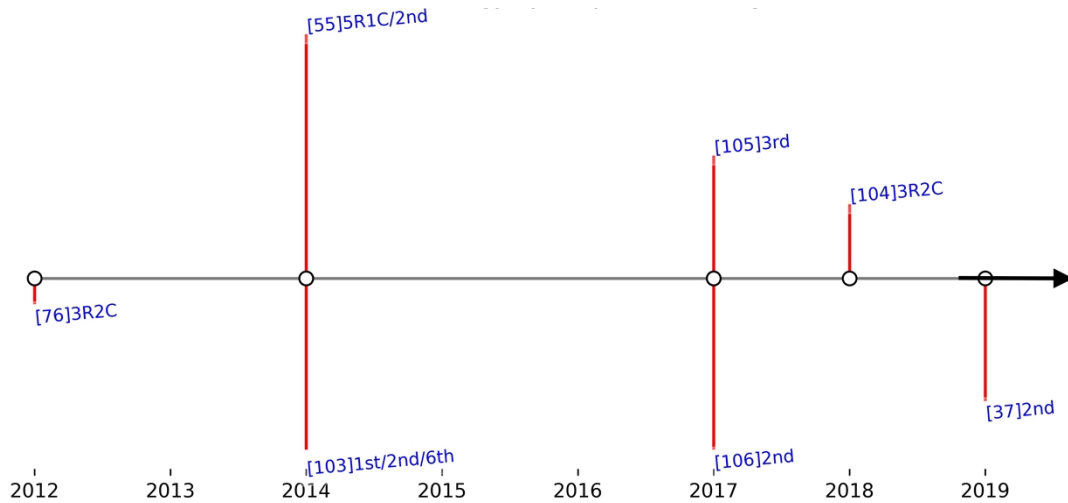


Figure 20. Timeline of district/urban energy by grey-box modeling.

This group of study can be classified into two groups based on the modeling platform:

(1) Modelica modeling language and (2) general modeling language.

(1) Modelica modeling language

Kim et al. proposed the reduced-order building models for urban scale energy simulation using Modelica. Their modeling achieved within 1% accuracy for annual energy consumption and 3% for hourly energy consumption [100]. A recent investigation was conducted for district heating networks coupled with buildings represented by 3R2C models in Modelica. The simulation results show such a simulation framework increased the system efficiency of geothermal-based district heating [101]. Kim et al. applied the 2nd-order model of buildings in the state-space format for district heating/cooling load calculations by Modelica. Their reduced-order models were able to achieve accuracy of 0.43% compared with reference buildings [37].

(2) General modeling language

Bueno et al. applied the 3R2C models to study the interactions between the building and the urban environment [68]. The models were validated against EnergyPlus data. They applied the model to explore the impacts of urban heat island effect on the building energy consumption coupling with an urban canopy model. The dominant factors are infiltration and ventilation. The authors also investigated the dominant mechanism through which the indoor environment might affect outdoor air temperatures, such as indoor air temperature, exfiltration heat, and waste heat from HVAC systems. Lauster et al. built the VDI 60071-1 models and compared it with ISO13790 models, both of which are thermal network models, in the city district scale. Their results showed that the VDI6007-1 model was good for city district energy use [46]. A study applied the 3rd-order RC models identified from the hourly electricity demand and weather measurements for an urban energy system. The urban heat island intensity was predicted to be around 2°C, with corresponding cooling load penalty to be 12% [102]. This study proposed an automatic simulation framework to generate urban-scale reduced-order building models. The generated models predicted the energy consumption with average errors of 10% compared with the reference models [103].

3.5 Building Grid Integration

Buildings account for 75% of the power demand in the United States[3]. Building-grid integration is an effective approach to facilitate the power demand reduction for buildings. The US Department of Energy issued reports about various aspects of grid-interactive efficient buildings, and pointed out multiple major requirements for buildings, including being efficient, connected, smart, and flexible [104]. Buildings are complicated in many aspects, such as having dynamic occupants with diverse behaviors. Furthermore, buildings have thermal inertia, which means their thermal dynamics are on the order of minutes or hours. While the grid has very short dynamic responses, on the order of 1 s (or less than 1 s). Building grid integration requires faster power consumption from buildings to match the time response of the grid. The RC building model is a good choice with fast energy calculations, especially in community or district scales. Multiple research efforts of using RC models have been applied to study the demand response issues. A summary is shown in Table 5. A timeline of RC modeling application in building-grid integration is shown in Figure 21, which shows that RC modeling has increasingly received more attention. However, the lack of unified software solution restrains its further application.

Table 5. Building grid integration.

RC model	RC parameter estimation method	Modeling scope	Single-zone or multi-zone	Note	Reference
4R2C, 6R2C, 6R3C, 7R3C	Identification	Power demand	Single		[35]
5th order	Identification	Power demand	Single	Smart grid	[105]
5R4C	Identification	Demand response	Single	Smart grid	[106]
1st order	Identification	Demand response	Single	Smart grid	[107]
2R1C	Identification	Demand response	Single	District grid	[108]
2nd order	Identification	Demand response	Single	Dynamic electricity pricing	[109]
5R2C	Identification	Demand response	Single		[45]

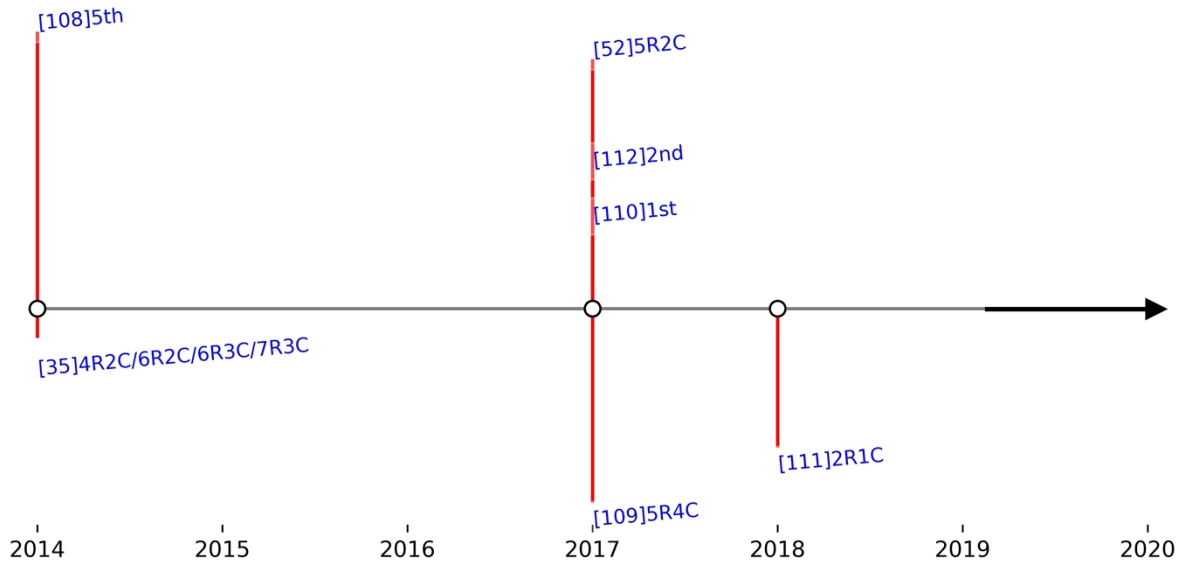


Figure 21. Timeline of building grid integration by grey-box modeling.

This group of applications can be categorized into (1) power demand prediction and (2) demand response analysis.

(1) Power demand prediction

An effort was conducted to develop and validate the multiple RC models for power demand of a single-zone office building based on TRNSYS data: 4R2C, 6R2C, 6R3C, and 7R3C. The results showed the 6R2C-based models had the best accuracy whereas the higher-order model did not improve the modeling accuracy [35]. A comparative study was conducted using five reduced-order models for a single-zone building. The purpose was to predict the power demand to better integrate with smart grid system. The results showed good prediction accuracy with reference data [105].

(2) Demand response analysis

Hu et al. applied the grey-box room thermal model (5R4C) to predict the indoor air temperature profile, which was used for demand response of a single-speed air-conditioner in smart grid integration [106]. Razmara et al. applied RC models for building and photovoltaic panels, and they used a battery to study the power controls in the demand response [107]. Mirakhorli and Dong developed MPC for residential buildings connected to the grid. The system included 15,000 residential building energy devices. For residential houses, the 2R1C models were applied [108]. The results concluded a 21% reduction in generation cost, 17% reduction in peak load, and reduced nodal voltage drop from the coordinated control system. Another study proposed a novel model-based optimal load scheduling method for residential air conditioners to implement automatic demand response according to day-ahead dynamic electricity prices. The RC models of residential buildings were identified. The developed framework worked in the programmable communicating thermostats and smart home energy manage systems [109]. Hu et al. studied the demand response potentials for a residential air-conditioning unit coupled with a room model (5R2C) [45]. The study showed a 26% power reduction for a typical summer day in Hong Kong.

As a summary for Section 3, the major applications of grey-box modeling are reviewed accordingly. The main takeaways of this section are that grey-box modeling (a) has a wide range of applications; (b) is successful in calculating the heat dynamics of envelope; (c) is faster in estimating thermal load without losing accuracy compared with white-box modeling; (d) is the best choice for controls and optimizations, whereas physical modeling is not robust for optimizations; (e) is suitable for district/urban-scale energy modeling because of the fast load calculation; and (f) is a good candidate for building-grid integration, which requires faster load calculation. A comprehensive summary is provided in Table 6.

Table 6. Building energy modeling summary by grey-box models.

(a) Heat dynamics analysis					
RC model	RC parameter	Variables studied	Envelope	Experiment or modeling	Reference
6th order	Identification	Zone temperature, heat flux	Radiant cooling/underfloor ventilation	Both	[47]
1st order	Identification	Surface temperature, heat flux	General wall	Modeling	[69]
3R2C	Identification	Heat transfer coefficient	Solid wall/cavity wall	Modeling	[23]
4R2C	Identification	Surface temperature, heat flux	Wall with phase change materials	Modeling	[70]
1R1C	Identification	Zone temperature	General wall	Modeling	[71]
4th order	Average value	Thermal conductivity, heat flux	Photovoltaic thermoelectric wall system	Both	[72]
9th order	Optimization	Heat flux	General walls, including concrete slabs	Both	[73]
2R1C	Identification	Heat flux	Pipe-embedded concrete radiant floor	Both	[20]
4R3C	Identification	Zone temperature	General wall for multi-zone	Modeling	[74]
5R1C	Identification	Zone temperature	General wall for single building	Modeling	[75]
2R1C	Identification	Zone temperature	General wall	Modeling	[76]
(b) Thermal load estimation					
RC model	RC parameter	Variables studied	Multi-zone or single-zone	Experiment or modeling	Reference
5R1C	Average value	Heating/cooling load hourly	Single	Modeling	[77]
2R2C, 3R2C, 4R2C, 5R2C	Identification	Thermal load	Single	Modeling	[78]
5R1C, 7R2C	Average value	Thermal load	Single	Modeling	[79]
5R1C	Average value	Thermal load	Single	Modeling	[80]
5R1C	Identification	Thermal load	Multi-zone	Modeling	[38]
3rd order	Identification	Thermal load and zone temperature	Single	Modeling	[30]
3R2C, 2R2C	Identification	Thermal load	Single	Modeling	[81]
1R1C, 2R2C, 3R3C	Identification	Heat flux and zone temperature	Single	Modeling	[82]

2nd, 3rd, 4th, 5th, 6th, 7th order	Identification	Heat capacity, thermal resistance, UA-value	Multi-zone	Both	[31]
1R1C, 3R2C, 4R2C, 8R3C	Optimization	Thermal load and zone temperature	Single	Modeling	[83]
3R3C, 2R1C	Optimization	Energy consumption	Multi-zone	Modeling	[84]
3rd order	Optimization	Energy consumption	Multi-zone	Modeling	[32]
2R2C	Identification	Energy consumption and zone temperature	Single	Modeling	[85]
3R2C, 2R2C	Identification	Thermal load	Single	Modeling	[25]
3R2C	Direct calculation	Thermal load and zone temperature	Multi-zone	Modeling	[24]
3R2C	Direct calculation	Thermal load and zone temperature	Single	Modeling	[26]
3R2C	Identification	Thermal load and zone temperature	Multi-zone	Modeling	[86]
3R2C	Identification	Thermal load and zone temperature	Single	Modeling	[87]
3R2C, 2R2C	Identification	Thermal load	Single	Modeling	[42]
40th, 14th order	Identification	Thermal load	Single	Modeling	[61]
2R2C	Identification	Thermal load	Single	Modeling	[44]
7th order	Direct calculation	Thermal load and zone temperature	Multi-zone	Modeling	[88]
3R2C	Optimization	Thermal load	Single	Modeling	[89]

(c) Building control and optimization

RC model	RC parameter estimation method	Control	Control variables	Modeling or experiment	Reference
3R3C, 2R1C	Identification	Thermostat setback	Zone temperature	Modeling	[84]
4th order	Identification	MPC	Energy consumption	Modeling	[90]
3R2C	Identification	MPC	Energy consumption	Modeling	[91]
1st order	Identification	MPC	Energy consumption	Both	[92]
2nd order	Identification	MPC and heuristic control	Energy consumption	Both	[93]
2R2C	Identification	MPC with feed- forward control	Energy consumption	Both	[94]
2R2C	Identification	Model-based control	Energy consumption and zone temperature	Both	[95]
3R2C	Identification	MPC	Energy consumption and zone temperature	Modeling	[96]

3R2C	Identification	MPC	Energy consumption and zone temperature	Both	[97]
5R4C	Identification	MPC	Energy consumption and zone temperature	Both	[98]
2nd order	Identification	MPC	Energy consumption and zone temperature	Both	[48]
2nd order	Identification	MPC	Energy consumption and zone temperature	Both	[99]
1R1C, 2R1C, 3R2C, 6R4C	Identification	MPC	Zone temperature	Both	[29]

(d) District/urban energy modeling

RC model	RC parameter estimation method	Modeling scope	Predicted variables	Note	Reference
3R2C	Identification	Urban	Energy consumption		[68]
5R1C, 2nd order	Identification	District	Energy consumption	ISO13790, VDI6007-1	[46]
1st, 2nd, 6th order	Identification	Urban	Energy consumption	Modelica	[100]
3R2C	Identification	District	Heating energy	Modelica	[101]
2nd order	Identification	District	Energy consumption	Modelica	[37]
3rd order	Identification	Urban	Energy consumption		[102]
2nd order	Identification	Urban	Energy consumption		[103]

(e) Building grid integration

RC model	RC parameter estimation method	Modeling scope	Single-zone or multi-zone	Note	Reference
4R2C, 6R2C, 6R3C, 7R3C	Identification	Power demand	Single		[35]
5th order	Identification	Power demand	Single	Smart grid	[105]
5R4C	Identification	Demand response	Single	Smart grid	[106]
1st order	Identification	Demand response	Single	Smart grid	[107]
2R1C	Identification	Demand response	Single	District grid	[108]
2nd order	Identification	Demand response	Single	Dynamic electricity pricing	[109]
5R2C	Identification	Demand response	Single		[45]

4. RC Modeling Toolbox

Multiple efforts have been made for RC modeling platform. They can be categorized into three groups: MATLAB, Modelica, and general-purpose free programming language (e.g., Python, C++). A summary is shown in Table 7.

(1) MATLAB

As a robust engineering programming platform, MATLAB has been widely used to create RC modeling packages. Kircher and Zhang developed BLDG [110], an open-source simplified building modeling toolbox for MPC, online estimation, and system identification. Similarly, Sturzenegger et al. developed BRCM [111], a toolbox for fast building thermodynamics simulation using RC models. However, those packages were designed for control purpose only.

(2) Modelica

Modelica, a dynamic-oriented modeling language, has been applied for RC modeling. FastBuildings is a Modelica-based reduce order building model [36] that was designed for supervisory controls and forecasting. Another effort is TEASER, a package for generating reduced-order models for urban-scale energy modeling [52]. However, those toolboxes rely on the Modelica environment, such as Dymola.

(3) General-purpose free programming

General-purpose programming languages have also been applied to build RC packages. RC_BuildingSimulator is a tool for an RC model-based building simulation program based on Python programming language, currently still under development [112]. The background of this package can be found in the literature [31,113,114]. The Argonne ISOmodel is another software tool under development by C++, implementing the ISO 13790 standard [33]. ISO 13790 was updated into ISO 52016 [34]. There were attempts developing Python version developed based on energy performance calculation tools [113, 117]. R language was also used to develop the grey-box modeling software CTSM-R [116].

Table 7. Grey-box building modeling software.

Name	Language	Notes	Reference
OpenBuild	MATLAB/Simulink		[117]
BLDG	MATLAB/Simulink		[110]
BRCM	MATLAB/Simulink		[111]
FastBuildings	Modelica	RC	[36]
RC_BuildingSimulator	Python	5R1C	[112]
Argonne ISOmodel	C++	ISO13790	[33]
Energy performance calculation	Excel/Python	Not available to public	[113]
CTSM-R	R	RC	[116]
TEASER	Python	Urban energy modeling	[52]
ISO 52016	Unknown		[34]

Overall, there is a lack of mature grey-box modeling tool that is user-friendly, generic, and robust with “plug and play” features. Such a software package would have a significant impact on the building energy domain.

5. Conclusions

RC models have various advantages across multiple sub-domains in building simulation. However, many hurdles prevent RC models from being adopted substantially. Therefore, a comprehensive overview is presented in this study.

First, this study reviews the fundamental aspects of grey-box models, including theoretical limitations, modeling of envelopes, modeling of zones and other building elements, diagram representation, and model order reduction. The two model creation approaches are also introduced in detail: forward and inverse. This paper aims to clarify various confusions of creating RC models.

Second, the applications of RC modeling in building domain are also summarized in detail, including heat dynamics, thermal load estimation, building control and optimization, district/urban energy modeling, and building grid integration. Thus, the RC model is a good choice for solving problems in the building energy domain.

Finally, the modeling software developed for reduced-order models are also summarized. Different software packages were developed based on multiple programming languages, including MATLAB, Python, Modelica, R, C++, and others. Because those software are either customized to individual research studies or have an extensive learning curve for general users (e.g., Modelica), a general building energy modeling software focusing grey-box modeling is desired.

Overall, grey-box modeling has good potential to solve the problems for which white-box models and black-box models are not suitable. For future studies, this study points out gaps of grey-box modeling. The following conclusions can be made: (1) Understanding the theoretical assumptions is important. (2) Following a standard naming convention, such as xRyC (e.g., 3R2C) for envelope, is ideal. (3) Individual researchers are left to choose the forward or inverse approach to create grey-box models (4) A need exists to investigate the order of grey-box modeling for efficient computations. (5) Grey-box modeling is available to an increasing number of applications, not only in energy/load estimation. (6) A user-friendly generic software package is needed to further promote energy modeling.

Acknowledgments

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. The authors declare that there is no conflict of interest.

References

- [1] DOE US. Buildings energy databook. Energy Efficiency & Renewable Energy Department 2011.
- [2] Conti JJ, Holtberg PD, Beamon JA, Schaal AM, Ayoub JC, Turnure JT. Annual energy outlook 2018. US Energy Information Administration 2018.
- [3] Sofos M, Langevin J, Deru M, Gupta E, Benne KS, Blum D, et al. Innovations in sensors and controls for building energy management: Research and development opportunities report for emerging technologies. National Renewable Energy Lab.(NREL), Golden, CO (United States); 2020.
- [4] Zero Net Energy 2016. <https://www.cpuc.ca.gov/zne/> (accessed March 20, 2021).
- [5] Afram A, Janabi-Sharifi F. Theory and applications of HVAC control systems–A review of model predictive control (MPC). *Building and Environment* 2014;72:343–55.
- [6] Torcellini P, Pless S, Deru M, Crawley D. Zero energy buildings: A critical look at the definition. National Renewable Energy Laboratory and Department of Energy, US 2006.
- [7] Li Y, O'Neill Z. An innovative fault impact analysis framework for enhancing building operations. *Energy and Buildings* 2019;199:311–31.
- [8] Li Y, O'Neill Z. A critical review of fault modeling of HVAC systems in buildings. *Build Simul* 2018;11:953–75. <https://doi.org/10.1007/s12273-018-0458-4>.
- [9] ASHRAE. ASHRAE Handbook: Fundamentals 2017.
- [10] About Building Energy Modeling, Department of Energy. <https://www.energy.gov/eere/buildings/about-building-energy-modeling> (accessed May 7, 2019).
- [11] Best Directory, Building Energy Software Tools <https://www.buildingenergysoftwaretools.com/> (accessed December 13, 2018).
- [12] EnergyPlus, EnergyPlus 2019. <https://energyplus.net/> (accessed January 29, 2019).
- [13] TRNSYS : Transient System Simulation Tool. <http://www.trnsys.com/> (accessed January 29, 2019).
- [14] Feng F, O'Neill Z. Identifying Models of HVAC Systems Using ARIMAX, 06/29-07/02, p. 3. 2020 ASHRAE Annual Virtual Conference (Online).
- [15] Zhang Y, O'Neill Z, Dong B, Augenbroe G. Comparisons of inverse modeling approaches for predicting building energy performance. *Building and Environment* 2015;86:177–90.
- [16] Dong B, Cao C, Lee SE. Applying support vector machines to predict building energy consumption in tropical region. *Energy and Buildings* 2005;37:545–53.
- [17] Zhao Y, Wen J, Xiao F, Yang X, Wang S. Diagnostic Bayesian networks for diagnosing air handling units faults – part I: Faults in dampers, fans, filters and sensors. *Applied Thermal Engineering* 2017;111:1272–86. <https://doi.org/10.1016/j.applthermaleng.2015.09.121>.
- [18] Zhao Y, Wen J, Wang S. Diagnostic Bayesian networks for diagnosing air handling units faults – Part II: Faults in coils and sensors. *Applied Thermal Engineering* 2015;90:145–57. <https://doi.org/10.1016/j.applthermaleng.2015.07.001>.
- [19] Li S, Wen J. A model-based fault detection and diagnostic methodology based on PCA method and wavelet transform. *Energy and Buildings* 2014;68:63–71.
- [20] Li A, Sun Y, Xu X. Development of a simplified resistance and capacitance (RC)-network model for pipe-embedded concrete radiant floors. *Energy and Buildings* 2017;150:353–75. <https://doi.org/10.1016/j.enbuild.2017.06.011>.
- [21] Hong T, Langevin J, Sun K. Building simulation: Ten challenges. *Build Simul* 2018;11:871–98. <https://doi.org/10.1007/s12273-018-0444-x>.

- [22] Kramer R, van Schijndel J, Schellen H. Simplified thermal and hygric building models: A literature review. *Frontiers of Architectural Research* 2012;1:318–25. <https://doi.org/10.1016/j.foar.2012.09.001>.
- [23] Gori V, Elwell CA. Estimation of thermophysical properties from in-situ measurements in all seasons: Quantifying and reducing errors using dynamic grey-box methods. *Energy and Buildings* 2018;167:290–300.
- [24] O'Neill Z, Narayanan S, Brahme R. Model-based thermal load estimation in buildings. *Proceedings of SimBuild* 2010;4:474–81.
- [25] Wang S, Xu X. Simplified building model for transient thermal performance estimation using GA-based parameter identification. *International Journal of Thermal Sciences* 2006;45:419–32. <https://doi.org/10.1016/j.ijthermalsci.2005.06.009>.
- [26] Chaturvedi N. An Inverse Gray-Box Model for Transient Building Load Prediction AU - Braun, James E. *HVAC&R Research* 2002;8:73–99. <https://doi.org/10.1080/10789669.2002.10391290>.
- [27] Goyal S, Barooah P. A method for model-reduction of non-linear thermal dynamics of multi-zone buildings. *Energy and Buildings* 2012;47:332–40.
- [28] Deng K, Barooah P, Mehta PG, Meyn SP. Building thermal model reduction via aggregation of states. *Proceedings of the 2010 American Control Conference, IEEE*; 2010, p. 5118–23.
- [29] Fux SF, Ashouri A, Benz MJ, Guzzella L. EKF based self-adaptive thermal model for a passive house. *Energy and Buildings* 2014;68:811–7. <https://doi.org/10.1016/j.enbuild.2012.06.016>.
- [30] Massa Gray F, Schmidt M. A hybrid approach to thermal building modelling using a combination of Gaussian processes and grey-box models. *Energy and Buildings* 2018;165:56–63. <https://doi.org/10.1016/j.enbuild.2018.01.039>.
- [31] Bacher P, Madsen H. Identifying suitable models for the heat dynamics of buildings. *Energy and Buildings* 2011;43:1511–22. <https://doi.org/10.1016/j.enbuild.2011.02.005>.
- [32] Giretti A, Vaccarini M, Casals M, Macarulla M, Fuertes A, Jones RV. Reduced-order modeling for energy performance contracting. *Energy and Buildings* 2018;167:216–30.
- [33] ANL hourly and monthly ISOmodel. Contribute to Argonne-National-Laboratory/ISOmodel development by creating an account on GitHub. Argonne National Laboratory; 2018.
- [34] Mazzarella L, Scoccia R, Colombo P, Motta M. Improvement to EN ISO 52016-1:2017 hourly heat transfer through a wall assessment: the Italian National Annex. *Energy and Buildings* 2020;210:109758. <https://doi.org/10.1016/j.enbuild.2020.109758>.
- [35] Berthou T, Stabat P, Salvazet R, Marchio D. Development and validation of a gray box model to predict thermal behavior of occupied office buildings. *Energy and Buildings* 2014;74:91–100.
- [36] De Coninck R, Magnusson F, \AAkesson J, Helsen L. Toolbox for development and validation of grey-box building models for forecasting and control. *Journal of Building Performance Simulation* 2016;9:288–303.
- [37] Kim E-J, He X, Roux J-J, Johannes K, Kuznik F. Fast and accurate district heating and cooling energy demand and load calculations using reduced-order modelling. *Applied Energy* 2019;238:963–71. <https://doi.org/10.1016/j.apenergy.2019.01.183>.

- [38] Bruno R, Pizzuti G, Arcuri N. The Prediction of Thermal Loads in Building by Means of the EN ISO 13790 Dynamic Model: A Comparison with TRNSYS. *Energy Procedia* 2016;101:192–9. <https://doi.org/10.1016/j.egypro.2016.11.025>.
- [39] Ogunsola OT, Song L. Review and Evaluation of Using R-C Thermal Modeling of Cooling Load Prediction for HVAC System Control Purpose 2012:735–43. <https://doi.org/10.1115/IMECE2012-86988>.
- [40] Ljung L. Black-box models from input-output measurements. IMTC 2001. Proceedings of the 18th IEEE instrumentation and measurement technology conference. Rediscovering measurement in the age of informatics (Cat. No. 01CH 37188), vol. 1, IEEE; 2001, p. 138–46.
- [41] Kircher KJ, Zhang KM. On the lumped capacitance approximation accuracy in RC network building models. *Energy and Buildings* 2015;108:454–62.
- [42] Wang S, Xu X. Parameter estimation of internal thermal mass of building dynamic models using genetic algorithm. *Energy Conversion and Management* 2006;47:1927–41.
- [43] Cui B, Fan C, Munk J, Mao N, Xiao F, Dong J, et al. A hybrid building thermal modeling approach for predicting temperatures in typical, detached, two-story houses. *Applied Energy* 2019;236:101–16. <https://doi.org/10.1016/j.apenergy.2018.11.077>.
- [44] Dewson T, Day B, Irving AD. Least squares parameter estimation of a reduced order thermal model of an experimental building. *Building and Environment* 1993;28:127–37. [https://doi.org/10.1016/0360-1323\(93\)90046-6](https://doi.org/10.1016/0360-1323(93)90046-6).
- [45] Hu M, Xiao F. Investigation of the demand response potentials of residential air conditioners using grey-box room thermal model. *Energy Procedia* 2017;105:2759–65.
- [46] Lauster M, Teichmann J, Fuchs M, Streblow R, Mueller D. Low order thermal network models for dynamic simulations of buildings on city district scale. *Building and Environment* 2014;73:223–31. <https://doi.org/10.1016/j.buildenv.2013.12.016>.
- [47] Zhang D, Xia X, Cai N. A dynamic simplified model of radiant ceiling cooling integrated with underfloor ventilation system. *Applied Thermal Engineering* 2016;106:415–22. <https://doi.org/10.1016/j.applthermaleng.2016.06.017>.
- [48] Hazyuk I, Ghiaus C, Penhouet D. Model Predictive Control of thermal comfort as a benchmark for controller performance. *Automation in Construction* 2014;43:98–109.
- [49] Gray FM, Schmidt M. Thermal building modelling using Gaussian processes. *Energy and Buildings* 2016;119:119–28.
- [50] M. H. Wit D, Driessen HH. ELAN - A computer model for building energy design. *Building and Environment* 1988;23:285–9. [https://doi.org/10.1016/0360-1323\(88\)90034-0](https://doi.org/10.1016/0360-1323(88)90034-0).
- [51] Hong T, Jiang Y. A new multizone model for the simulation of building thermal performance. *Building and Environment* 1997;32:123–8.
- [52] Full article: TEASER: an open tool for urban energy modelling of building stocks. <https://www.tandfonline.com/doi/full/10.1080/19401493.2017.1283539> (accessed June 27, 2020).
- [53] Andriamamonjy A, Klein R, Saelens D. Automated grey box model implementation using BIM and Modelica. *Energy and Buildings* 2019.
- [54] Chiuso A, Pillonetto G. System identification: A machine learning perspective. *Annual Review of Control, Robotics, and Autonomous Systems* 2019;2:281–304.
- [55] Nelles O. Nonlinear system identification. IOP Publishing; 2002.
- [56] Unbehauen H, Rao GP. A review of identification in continuous-time systems. *Annual Reviews in Control* 1998;22:145–71.

- [57] Okuyama H, Onishi Y. System parameter identification theory and uncertainty analysis methods for multi-zone building heat transfer and infiltration. *Building and Environment* 2012;54:39–52.
- [58] Kim D, Cai J, Ariyur KB, Braun JE. System identification for building thermal systems under the presence of unmeasured disturbances in closed loop operation: Lumped disturbance modeling approach. *Building and Environment* 2016;107:169–80. <https://doi.org/10.1016/j.buildenv.2016.07.007>.
- [59] Antoulas AC, Sorensen DC, Gugercin S. A survey of model reduction methods for large-scale systems. 2000.
- [60] Antoulas AC. Approximation of Large-scale Dynamical Systems. SIAM; 2005.
- [61] Gouda MM, Danaher S, Underwood CP. Building thermal model reduction using nonlinear constrained optimization. *Building and Environment* 2002;37:1255–65.
- [62] Lucas T. Linear system reduction by impulse energy approximation. *IEEE Transactions on Automatic Control* 1985;30:784–6.
- [63] Kim D, Braun JE. A general approach for generating reduced-order models for large multi-zone buildings. *Journal of Building Performance Simulation* 2015;8:435–48.
- [64] Rabenstein R. Application of model reduction techniques to building energy simulation. *Solar Energy* 1994;53:289–99.
- [65] Deng K, Goyal S, Barooah P, Mehta PG. Structure-preserving model reduction of nonlinear building thermal models. *Automatica* 2014;50:1188–95.
- [66] Ménézo C, Roux JJ, Virgone J. Modelling heat transfers in building by coupling reduced-order models. *Building and Environment* 2002;37:133–44.
- [67] Kramer R, van Schijndel J, Schellen H. Inverse modeling of simplified hygrothermal building models to predict and characterize indoor climates. *Building and Environment* 2013;68:87–99.
- [68] Bueno B, Norford L, Pigeon G, Britter R. A resistance-capacitance network model for the analysis of the interactions between the energy performance of buildings and the urban climate. *Building and Environment* 2012;54:116–25. <https://doi.org/10.1016/j.buildenv.2012.01.023>.
- [69] Jara EÁR, de la Flor FJS, Domínguez SÁ, Félix JLM, Lissén JMS. A new analytical approach for simplified thermal modelling of buildings: Self-Adjusting RC-network model. *Energy and Buildings* 2016;130:85–97.
- [70] Gao J, Yan T, Xu T, Ling Z, Wei G, Xu X. Development and experiment validation of variable-resistance-variable-capacitance dynamic simplified thermal models for shape-stabilized phase change material slab. *Applied Thermal Engineering* 2019;146:364–75. <https://doi.org/10.1016/j.applthermaleng.2018.09.124>.
- [71] Park H, Ruellan M, Bouvet A, Monmasson E, Bennacer R. Thermal parameter identification of simplified building model with electric appliance. *Electrical Power Quality and Utilisation (EPQU)*, 2011 11th International Conference on, IEEE; 2011, p. 1–6.
- [72] Luo Y, Zhang L, Liu Z, Wang Y, Meng F, Wu J. Thermal performance evaluation of an active building integrated photovoltaic thermoelectric wall system. *Applied Energy* 2016;177:25–39.
- [73] Weber T, Jóhannesson G. An optimized RC-network for thermally activated building components. *Building and Environment* 2005;40:1–14.

- [74] Bagheri A, Feldheim V, Thomas D, Ioakimidis CS. The adjacent walls effects in simplified thermal model of buildings. *Energy Procedia* 2017;122:619–24. <https://doi.org/10.1016/j.egypro.2017.07.359>.
- [75] Panão MJO, Santos CA, Mateus NM, da Graça GC. Validation of a lumped RC model for thermal simulation of a double skin natural and mechanical ventilated test cell. *Energy and Buildings* 2016;121:92–103.
- [76] Mathews EH, Richards PG, Lombard C. A first-order thermal model for building design. *Energy and Buildings* 1994;21:133–45.
- [77] Michalak P. The simple hourly method of EN ISO 13790 standard in Matlab/Simulink: A comparative study for the climatic conditions of Poland. *Energy* 2014;75:568–78. <https://doi.org/10.1016/j.energy.2014.08.019>.
- [78] Ogunsola OT, Song L. Application of a simplified thermal network model for real-time thermal load estimation. *Energy and Buildings* 2015;96:309–18.
- [79] Vivian J, Zarrella A, Emmi G, De Carli M. An evaluation of the suitability of lumped-capacitance models in calculating energy needs and thermal behaviour of buildings. *Energy and Buildings* 2017;150:447–65. <https://doi.org/10.1016/j.enbuild.2017.06.021>.
- [80] Michalak P. The development and validation of the linear time varying Simulink-based model for the dynamic simulation of the thermal performance of buildings. *Energy and Buildings* 2017;141:333–40.
- [81] Zhou Q, Wang S, Xu X, Xiao F. A grey-box model of next-day building thermal load prediction for energy-efficient control. *International Journal of Energy Research* 2008;32:1418–31. <https://doi.org/10.1002/er.1458>.
- [82] Rouchier S, Rabouille M, Oberlé P. Calibration of simplified building energy models for parameter estimation and forecasting: Stochastic versus deterministic modelling. *Building and Environment* 2018;134:181–90.
- [83] Harb H, Boyanov N, Hernandez L, Streblow R, Müller D. Development and validation of grey-box models for forecasting the thermal response of occupied buildings. *Energy and Buildings* 2016;117:199–207. <https://doi.org/10.1016/j.enbuild.2016.02.021>.
- [84] Rosin S. Reduced Order Modeling for Virtual Building Commissioning 2018.
- [85] Coffman AR, Barooah P. Simultaneous identification of dynamic model and occupant-induced disturbance for commercial buildings. *Building and Environment* 2018;128:153–60.
- [86] Lee K, Braun JE. Model-based demand-limiting control of building thermal mass. *Building and Environment* 2008;43:1633–46. <https://doi.org/10.1016/j.buildenv.2007.10.009>.
- [87] Danza L, Belussi L, Meroni I, Salamone F, Floreani F, Piccinini A, et al. A simplified thermal model to control the energy fluxes and to improve the performance of buildings. *Energy Procedia* 2016;101:97–104.
- [88] Weedy BM, Isgar P. Model for the computation of transient temperatures in buildings. *Energy Conversion and Management* 1985;25:57–63.
- [89] Harish VSKV, Kumar A. Reduced order modeling and parameter identification of a building energy system model through an optimization routine. *Applied Energy* 2016;162:1010–23. <https://doi.org/10.1016/j.apenergy.2015.10.137>.
- [90] Li S, Joe J, Hu J, Karava P. System identification and model-predictive control of office buildings with integrated photovoltaic-thermal collectors, radiant floor heating and active thermal storage. *Solar Energy* 2015;113:139–57.

- [91] Hu J, Karava P. A state-space modeling approach and multi-level optimization algorithm for predictive control of multi-zone buildings with mixed-mode cooling. *Building and Environment* 2014;80:259–73.
- [92] Huang H, Chen L, Hu E. A new model predictive control scheme for energy and cost savings in commercial buildings: An airport terminal building case study. *Building and Environment* 2015;89:203–16.
- [93] Hu J, Karava P. Model predictive control strategies for buildings with mixed-mode cooling. *Building and Environment* 2014;71:233–44.
- [94] Zhuang J, Chen Y, Chen X. A new simplified modeling method for model predictive control in a medium-sized commercial building: a case study. *Building and Environment* 2018;127:1–12.
- [95] Kui S, Wang J, Maomao HU, GAO D. A model-based control strategy to recover cooling energy from thermal mass in commercial buildings. *Energy* 2019.
- [96] Salakij S, Yu N, Paolucci S, Antsaklis P. Model-Based Predictive Control for building energy management. I: Energy modeling and optimal control. *Energy and Buildings* 2016;133:345–58.
- [97] Yang S, Wan MP, Ng BF, Zhang T, Babu S, Zhang Z, et al. A state-space thermal model incorporating humidity and thermal comfort for model predictive control in buildings. *Energy and Buildings* 2018;170:25–39.
- [98] Viot H, Sempey A, Mora L, Batsale JC, Malvestio J. Model predictive control of a thermally activated building system to improve energy management of an experimental building: Part I—Modeling and measurements. *Energy and Buildings* 2018;172:94–103.
- [99] Zong Y, Böning GM, Santos RM, You S, Hu J, Han X. Challenges of implementing economic model predictive control strategy for buildings interacting with smart energy systems. *Applied Thermal Engineering* 2017;114:1476–86.
- [100] Kim E-J, Plessis G, Hubert J-L, Roux J-J. Urban energy simulation: Simplification and reduction of building envelope models. *Energy and Buildings* 2014;84:193–202.
- [101] Fabre A, Thomas R, Duplessis B, Tran C-T, Stabat P. Dynamic modeling for evaluation of triple-pipe configuration potential in geothermal district heating networks. *Energy Conversion and Management* 2018;173:461–9.
- [102] Afshari A, Liu N. Inverse modeling of the urban energy system using hourly electricity demand and weather measurements, Part 2: Gray-box model. *Energy and Buildings* 2017;157:139–56. <https://doi.org/10.1016/j.enbuild.2017.01.052>.
- [103] Heidarinejad M, Mattise N, Dahlhausen M, Sharma K, Benne K, Macumber D, et al. Demonstration of reduced-order urban scale building energy models. *Energy and Buildings* 2017;156:17–28.
- [104] Grid-Interactive Efficient Buildings. EnergyGov. <https://www.energy.gov/eere/buildings/grid-interactive-efficient-buildings> (accessed July 11, 2020).
- [105] Reynders G, Diriken J, Saelens D. Quality of grey-box models and identified parameters as function of the accuracy of input and observation signals. *Energy and Buildings* 2014;82:263–74. <https://doi.org/10.1016/j.enbuild.2014.07.025>.
- [106] Hu M, Xiao F, Wang L. Investigation of demand response potentials of residential air conditioners in smart grids using grey-box room thermal model. *Applied Energy* 2017;207:324–35. <https://doi.org/10.1016/j.apenergy.2017.05.099>.

- [107] Razmara M, Bharati GR, Hanover D, Shahbakhti M, Paudyal S, Robinett RD. Building-to-grid predictive power flow control for demand response and demand flexibility programs. *Applied Energy* 2017;203:128–41.
<https://doi.org/10.1016/j.apenergy.2017.06.040>.
- [108] Mirakhorli A, Dong B. Model predictive control for building loads connected with a residential distribution grid. *Applied Energy* 2018;230:627–42.
<https://doi.org/10.1016/j.apenergy.2018.08.051>.
- [109] Hu M, Xiao F. Model-based optimal load control of inverter-driven air conditioners responding to dynamic electricity pricing. *Energy Procedia* 2017;142:1953–9.
- [110] Kircher KJ, Zhang KM. Testing building controls with the BLDG toolbox. 2016 American Control Conference (ACC), 2016, p. 1472–7.
<https://doi.org/10.1109/ACC.2016.7525124>.
- [111] Sturzenegger D, Gyalistras D, Semeraro V, Morari M, Smith RS. BRCM Matlab Toolbox: Model generation for model predictive building control. 2014 American Control Conference, 2014, p. 1063–9. <https://doi.org/10.1109/ACC.2014.6858967>.
- [112] RC Model of a Building. Contribute to architecture-building-systems/RC_BuildingSimulator development by creating an account on GitHub. Architecture and Building Systems; 2019.
- [113] Madsen H, Holst J. Estimation of continuous-time models for the heat dynamics of a building. *Energy and Buildings* 1995;22:67–79.
- [114] Sonderegger R. Diagnostic tests determining the thermal response of a house 1977.
- [115] Chopson P. A project to convert the Excel based Energy Performance Calculator to Python: pchopson/EPC-To-Python. 2019.
- [116] Madsen H. CTSM-R 2019. <http://henrikmadsen.org/software/ctsm-r/>.
- [117] Gorecki TT, Qureshi FA, Jones CN. OpenBuild: An integrated simulation environment for building control. 2015 IEEE Conference on Control Applications (CCA), IEEE; 2015, p. 1522–7.
- [118] EPC Integration, Georgia Institute of Technology, Atlanta, GA.
<http://imagine.gatech.edu/node/511> (accessed March 4, 2019).